Modeling and Analysis of Chladni Plate Nodal Patterns

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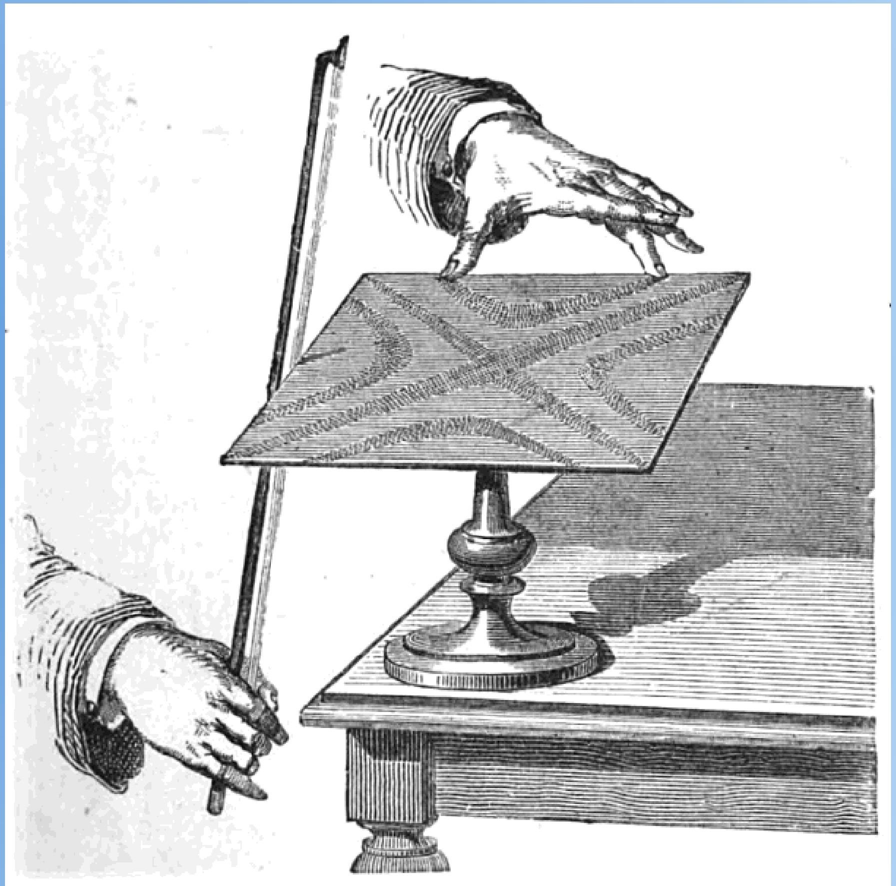
Overview

- Historical Context
- Theoretical Model
- Data Collection
- Image Processing
- Qualitative Comparison
- Conclusions

Historical Implementation

- Mounted plate
- Driven from side
- Induce vibrations with bow
- Elusive physical model



Refining the model

- Problem successfully modeled (Sophie Germain, 1776-1831)
- Improved understanding of boundary conditions
- Fourth order PDE with cross-terms
- We will model off-center driven behavior

Kirchhoff-Love Equation

$$\underbrace{D\,\nabla^4\,w}_{\text{Stiffness Term}} + \underbrace{\rho\,h\,\frac{\partial^2 w}{\partial t^2}}_{\text{Inertial Term}} = \underbrace{S(t,x)}_{\text{Sourcing Term}}$$

$$D = \frac{E\,h^3}{12(1-\nu^2)} \quad \Rightarrow$$

E is Young's Modulus

ν is Poisson's ratio

ρ is plate density

h is plate thickness

$$\nabla^4\,w = \frac{\partial^4 w}{\partial x^4} + \frac{\partial^4 w}{\partial x^2 \partial y^2} + \frac{\partial^4 w}{\partial y^4}$$

Assumptions

- Ideal plate
- Uniform mass density
- Minimal bending
- Free boundary

Frequency Domain

$$w(x, y, t) = u(x, y)e^{i\omega t}$$

$$(e^{i\omega t}\, D)\nabla^4\, u - \left(\rho h\, \omega^2 e^{i\omega t}\right)u = f(x)\, e^{i\omega t}$$

$$D\nabla^4\, u - \rho h\, \omega^2 u = f(x)$$

$$\frac{\partial u}{\partial x} = 0 \qquad \frac{\partial^2 u}{\partial x^2} = 0 \qquad \frac{\partial^3 u}{\partial x^3} = 0$$

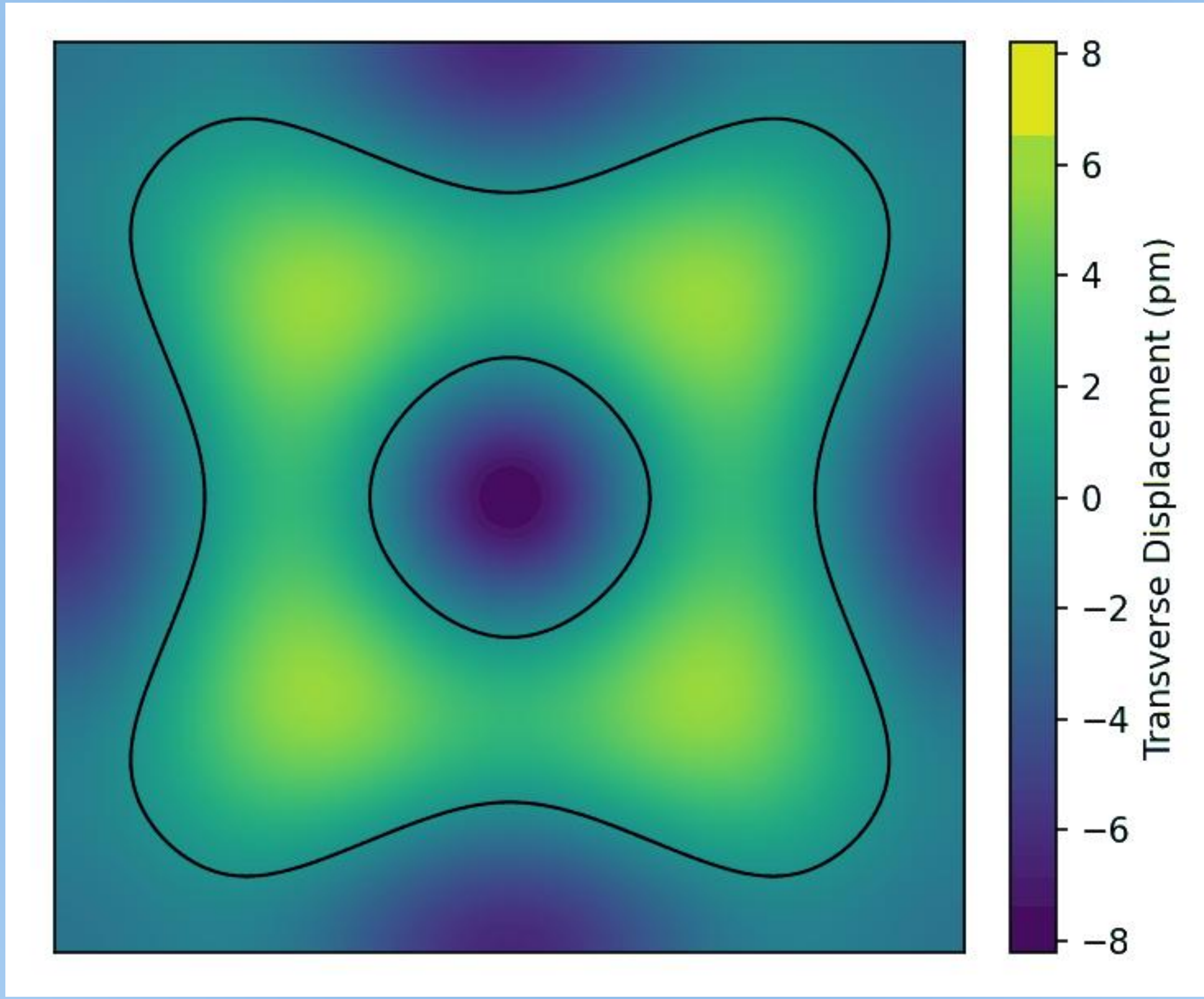
Discretize

- Operators go to finite difference quotients

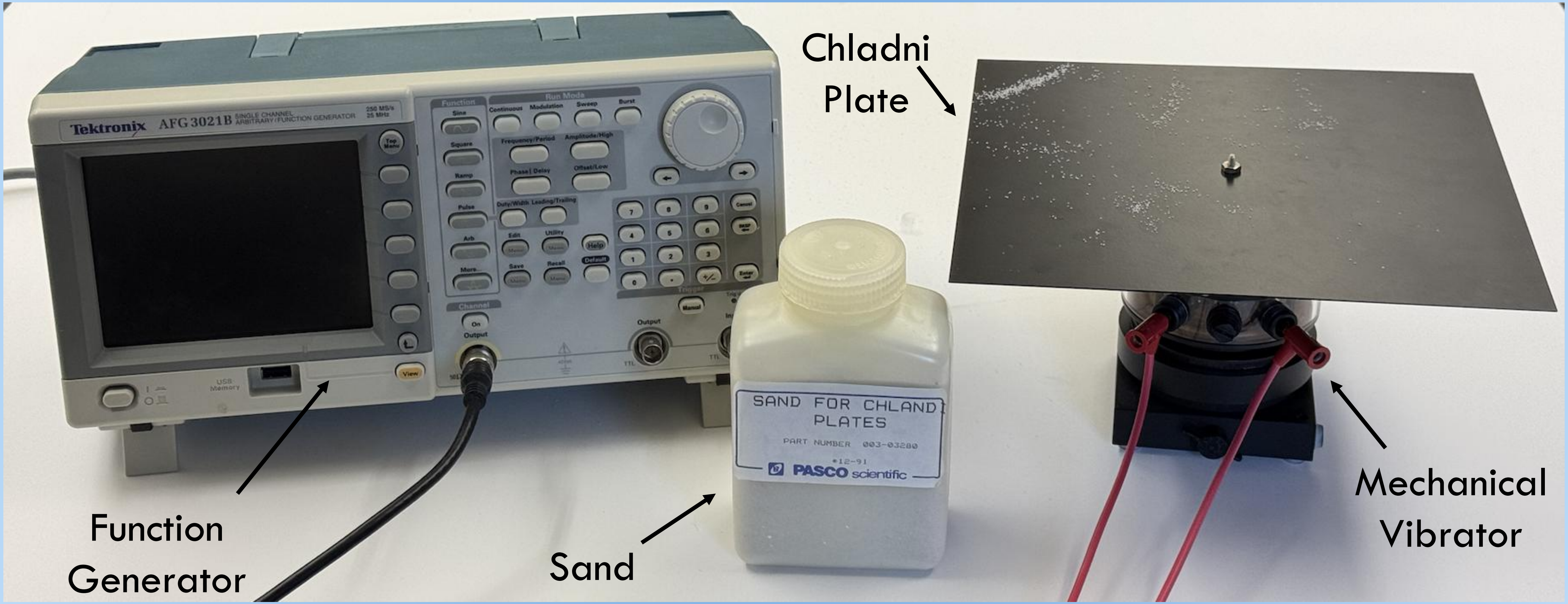
$$\sum_j m_{ij} u_j + (\rho h)\omega^2 u_i = f_i(x)$$

$$M\vec{u} + (\rho h)\omega^2 \vec{u} = \vec{f}$$

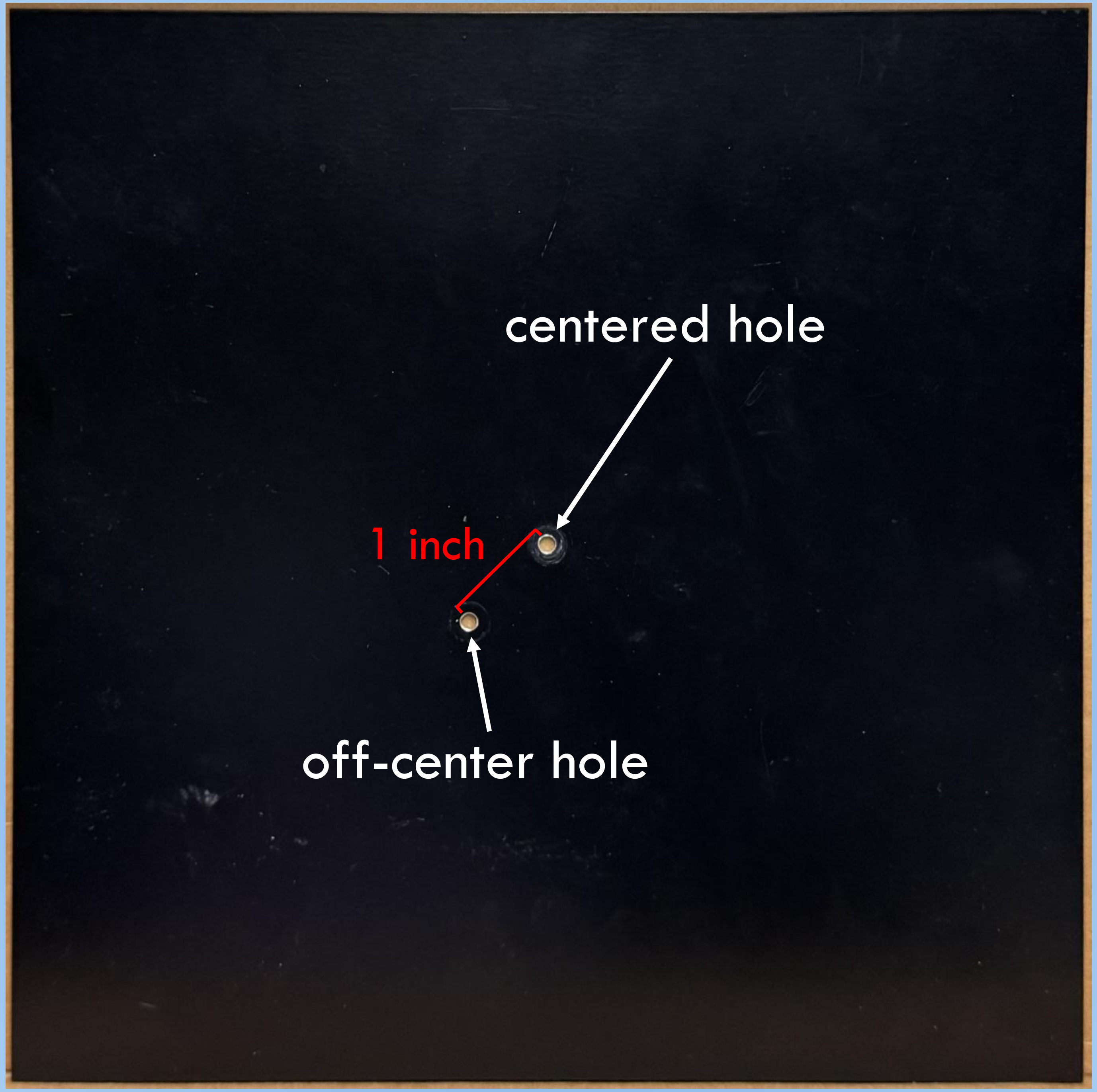
Standing Wave Behavior



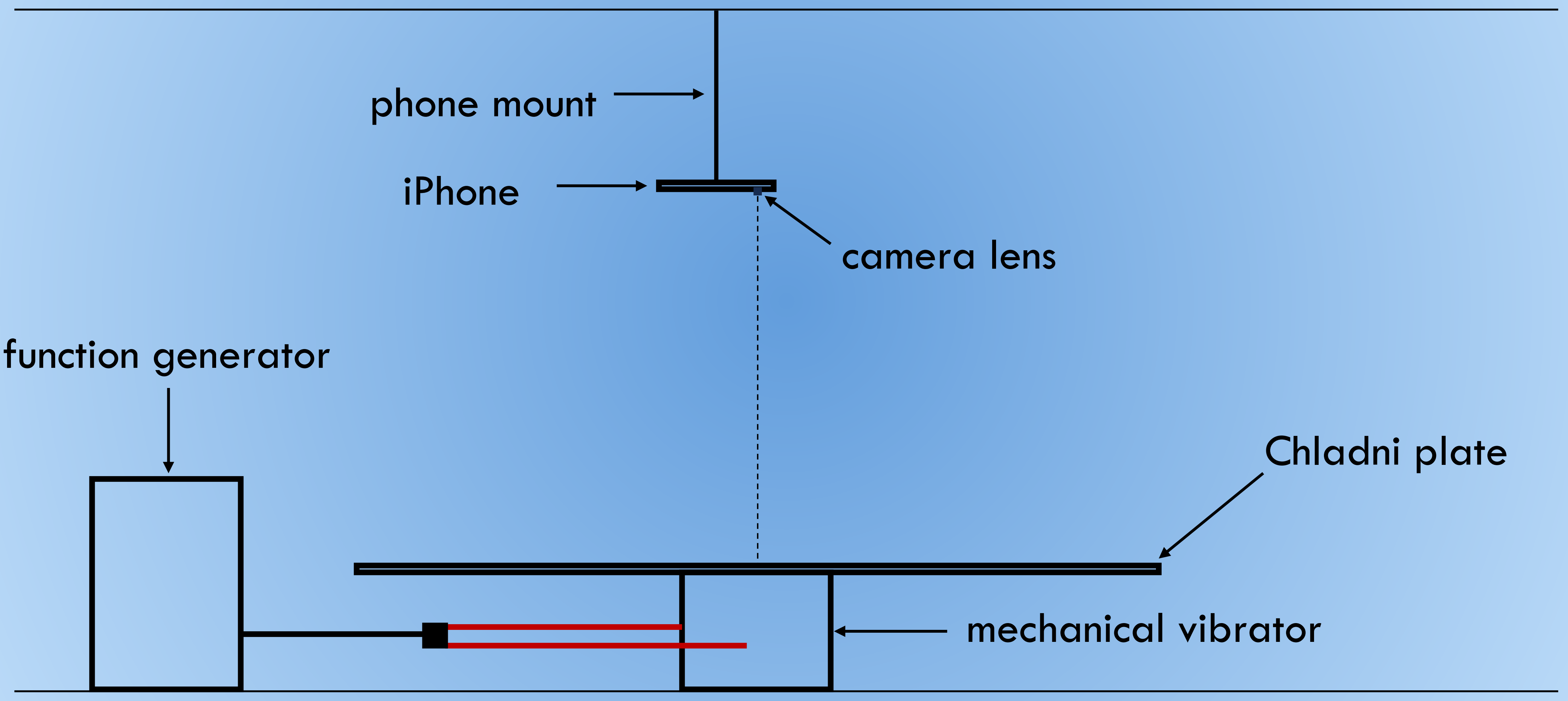
Materials


Chladni
Plate
Tektronix AFG 3021B
Mechanical
Vibrator
SAND FOR CHLANDI
PLATES
PASCO scientific
Function
Generator
Sand

Square Plate



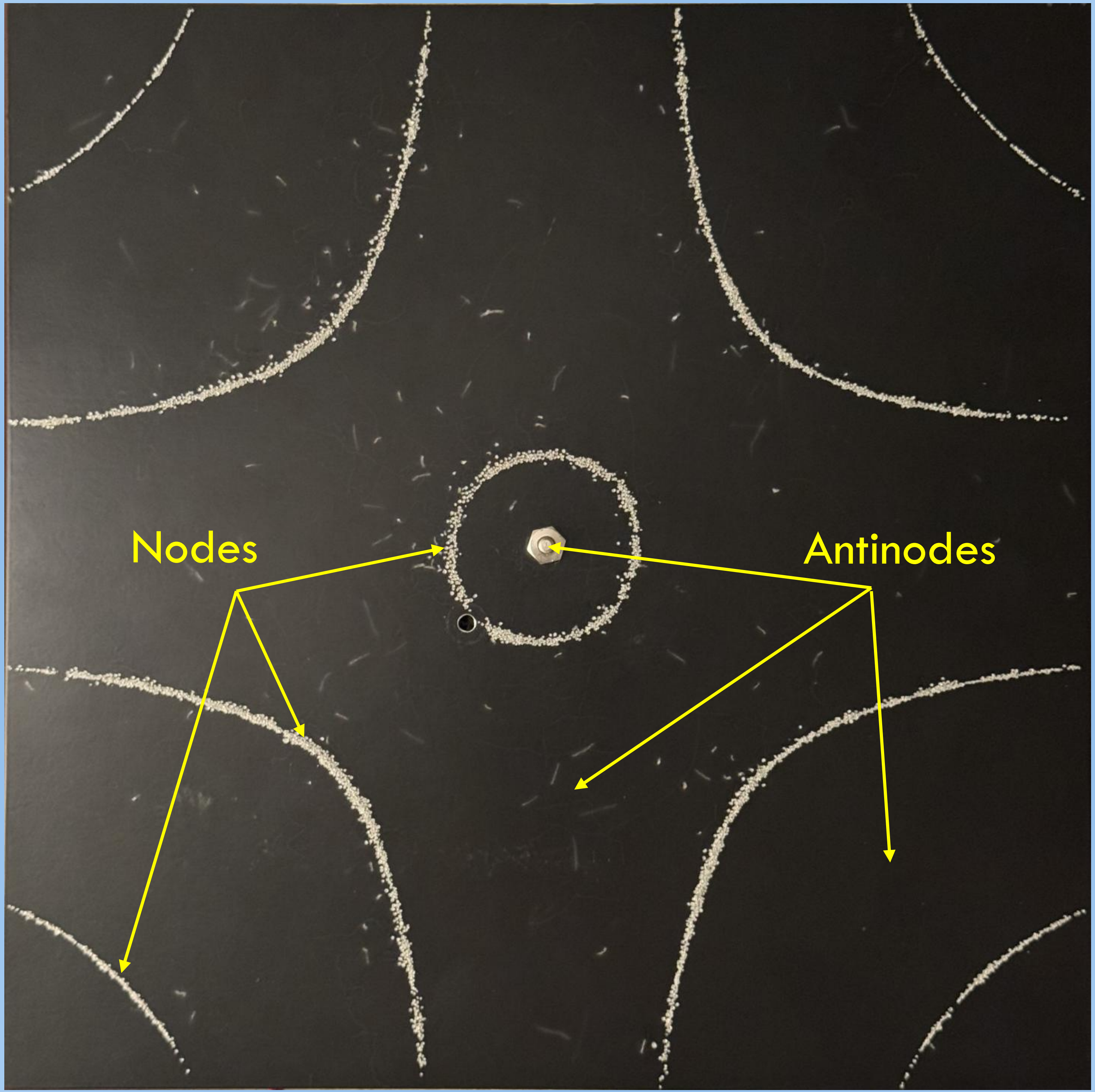
Apparatus


phone mount
iPhone
camera lens
function generator
Chladni plate
mechanical vibrator

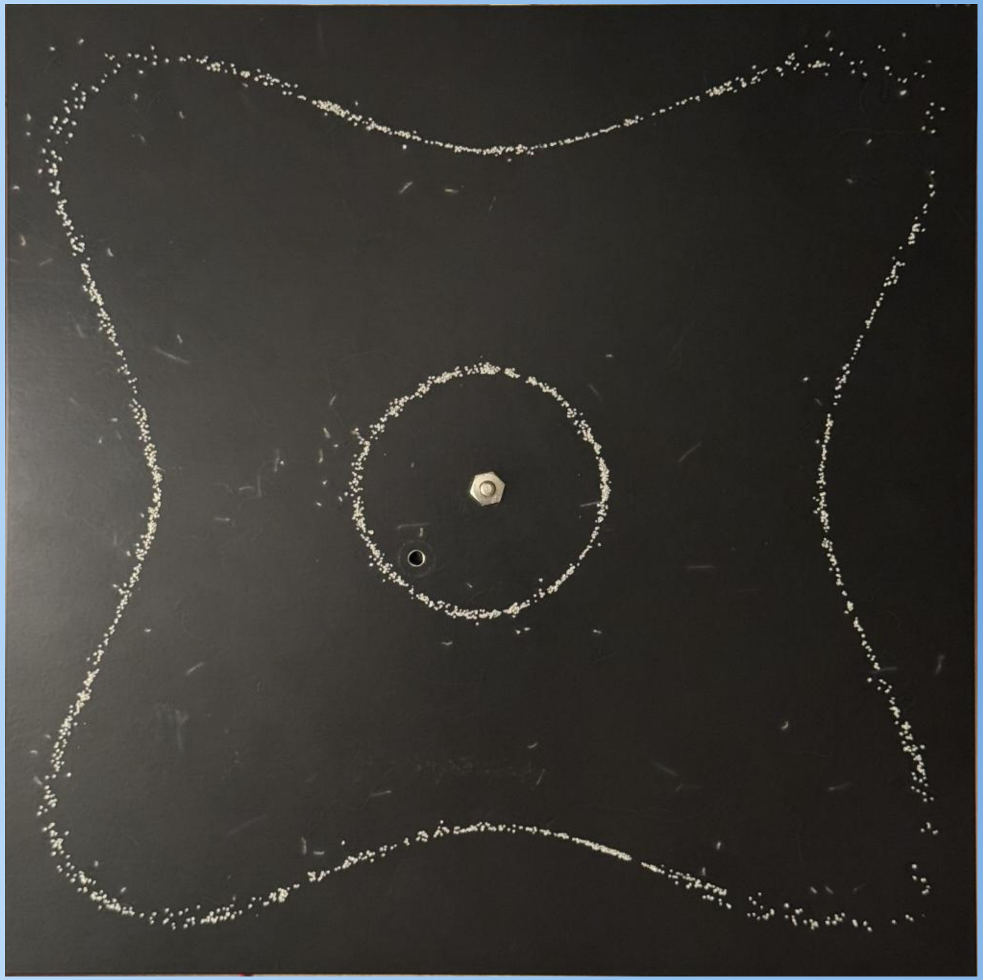
Data Collection

- Sprinkle sand on plate
- Drive plate at resonant frequency
- Allow sand to settle at nodes
- Take photo

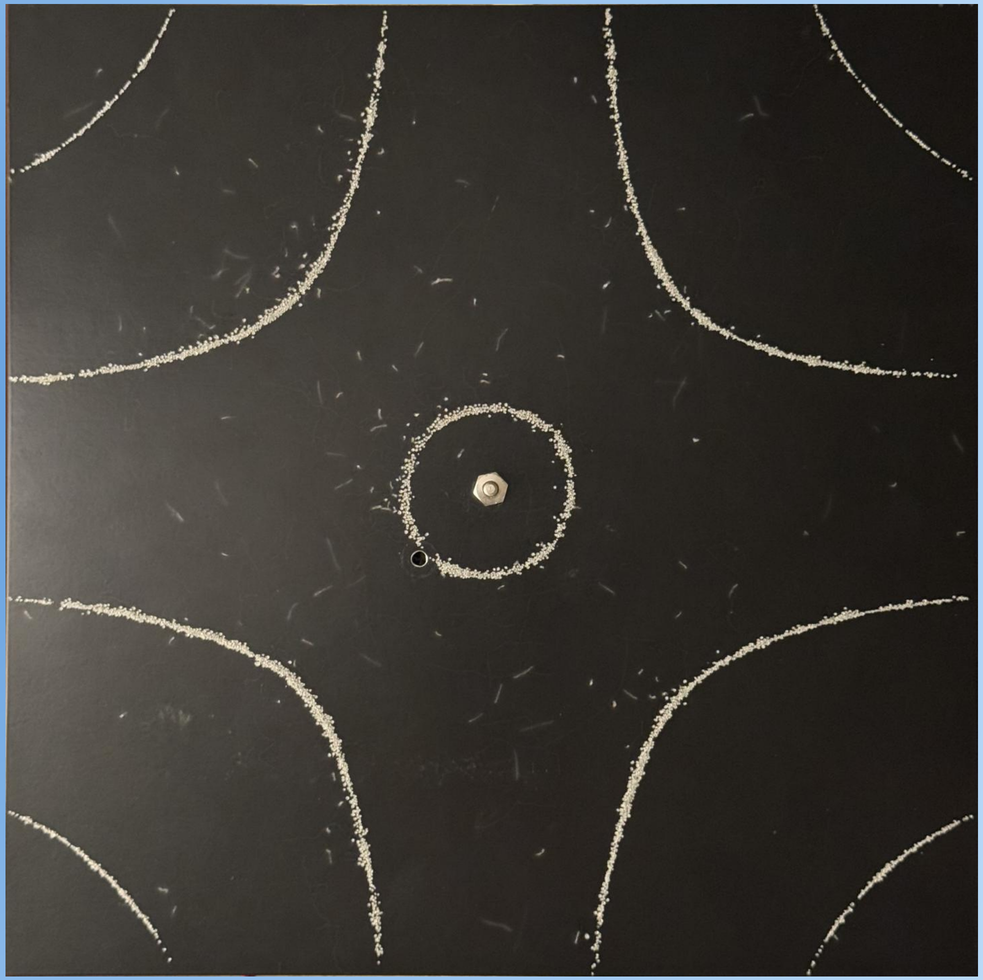
Nodal Lines



Raw Images: Center Driven

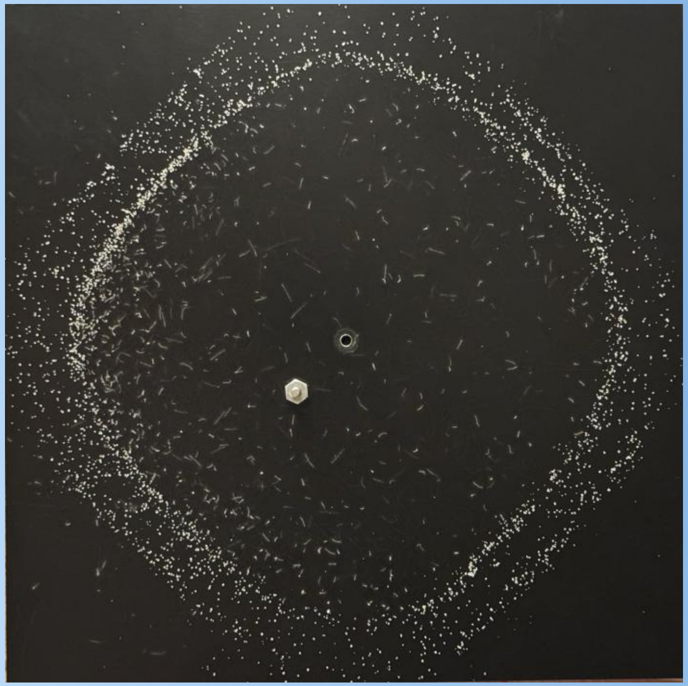


313 Hz

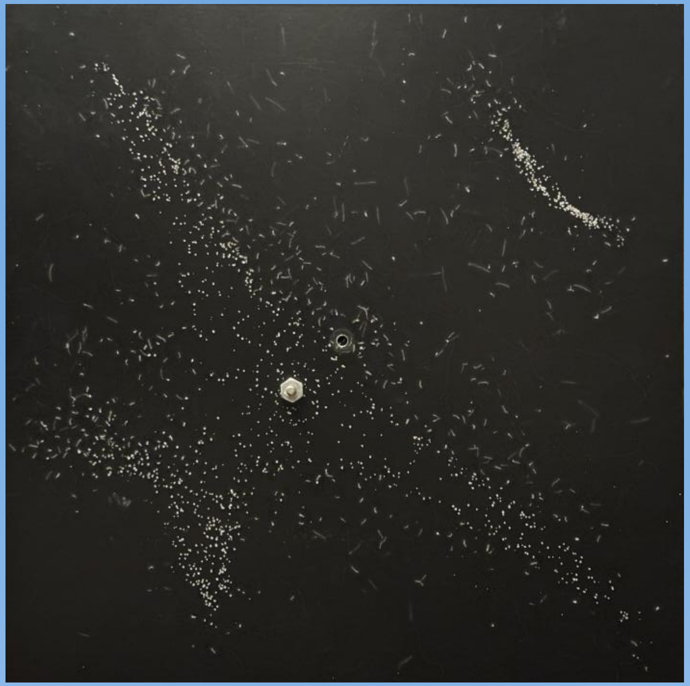


455 Hz

Raw Images:
Off-Center Driven



86 Hz



106 Hz

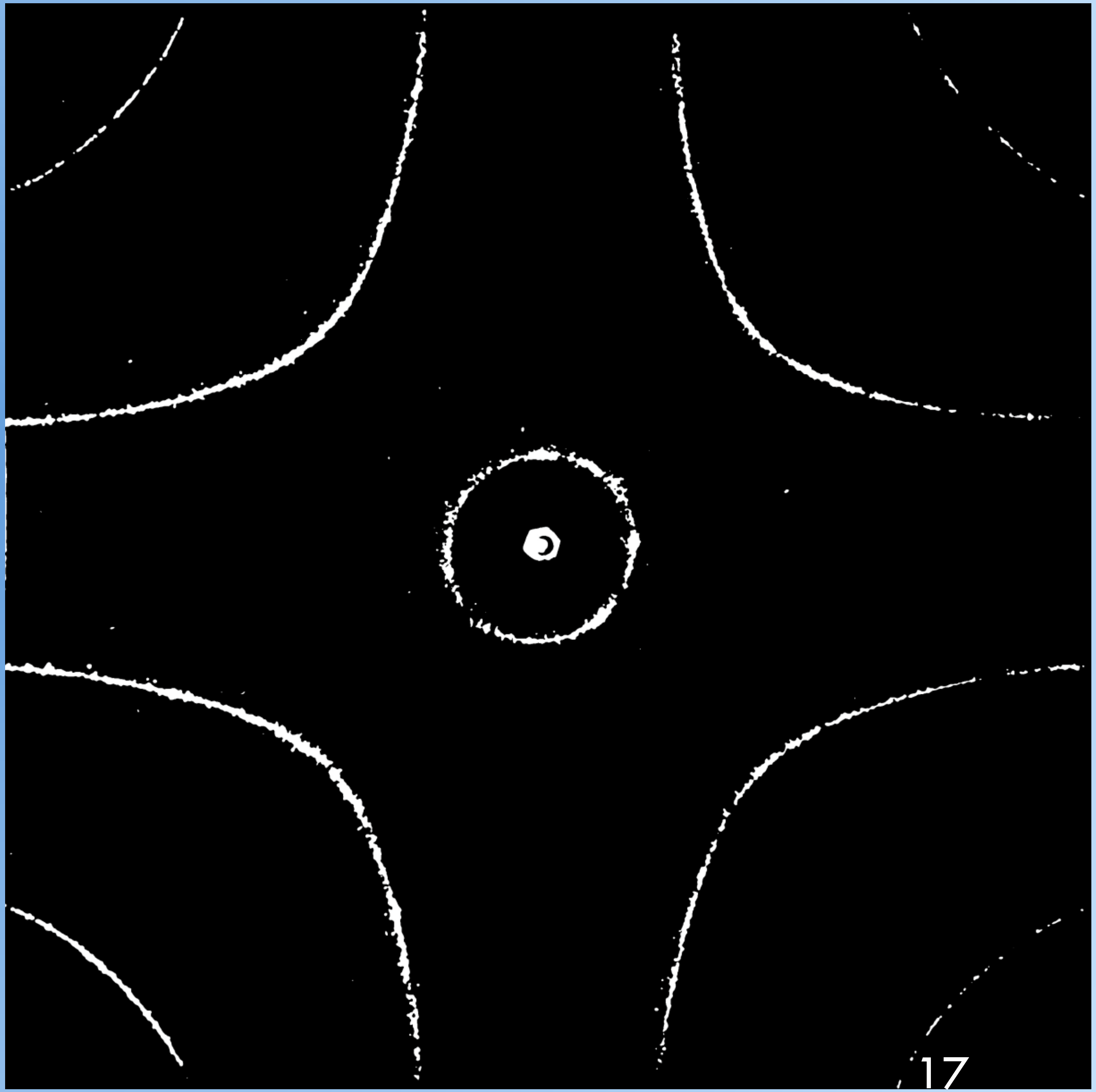
241Hz

Converting Images to Binary Arrays

Input image into Python code that outputs an array of 1s and 0s

- Apply Gaussian blur to image
- Convert RGB to grayscale
- Determine and apply appropriate threshold for white
- Populate into an array

Re-render binary array as image



Combining Binary Arrays

- Combine several images at given frequency
  - 0 everywhere all files have a 0, 1 otherwise
- Generate a composite binary array for each frequency

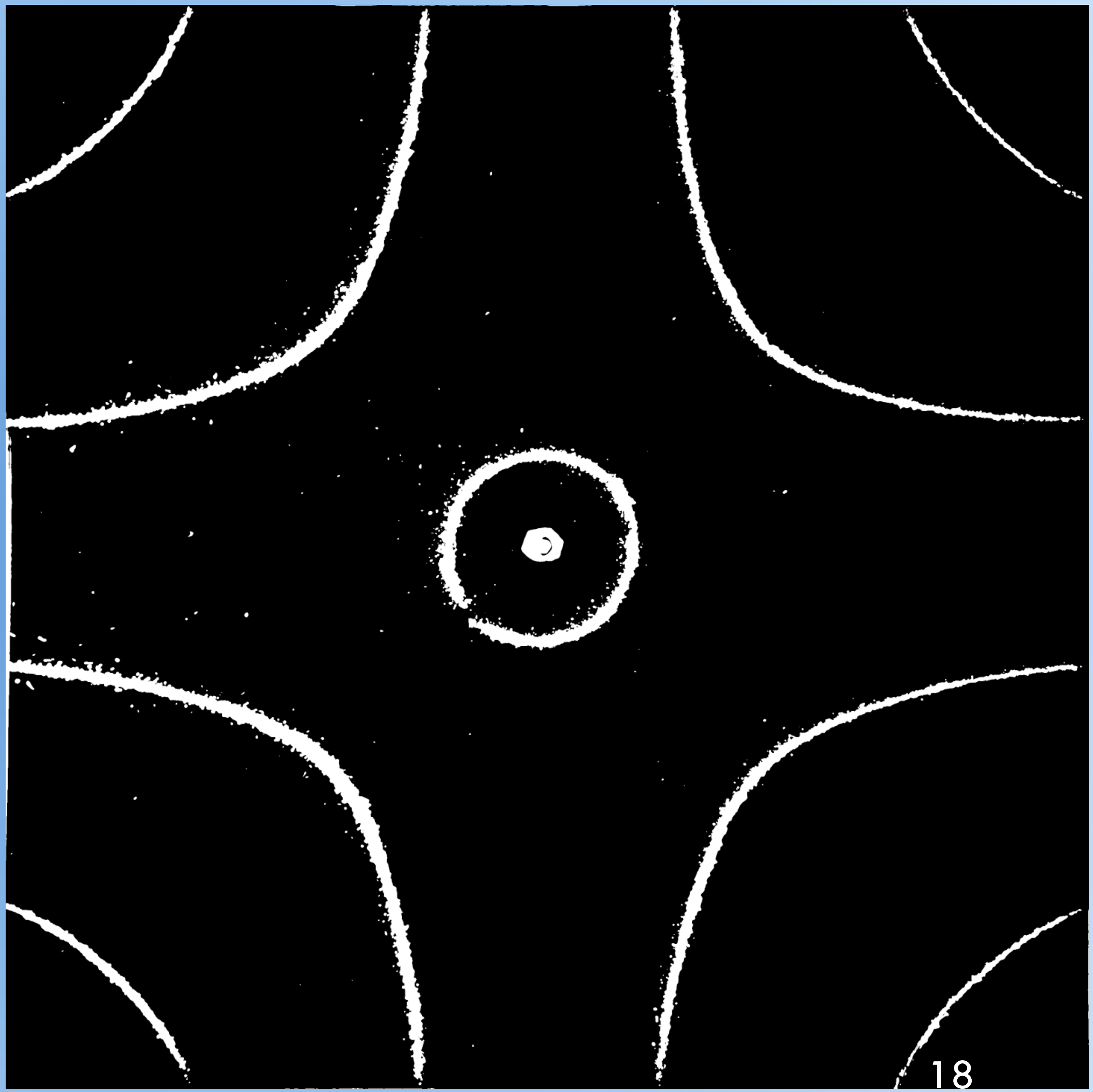
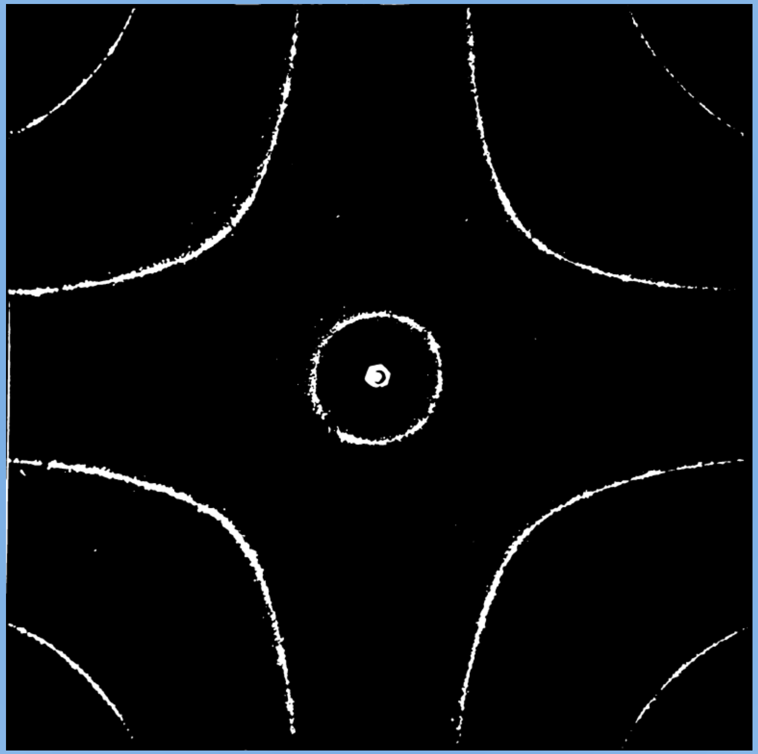
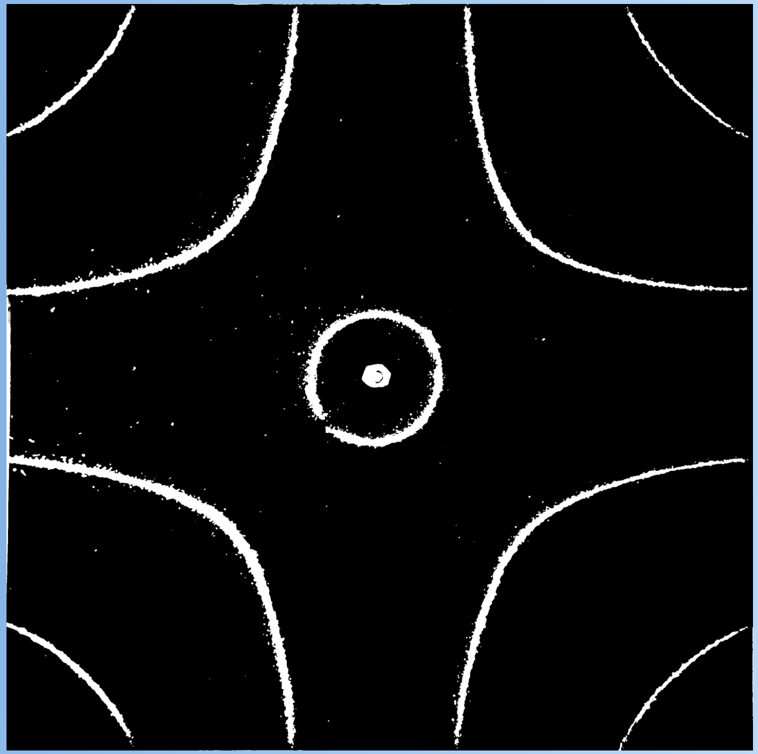


Image Processing

Raw Image



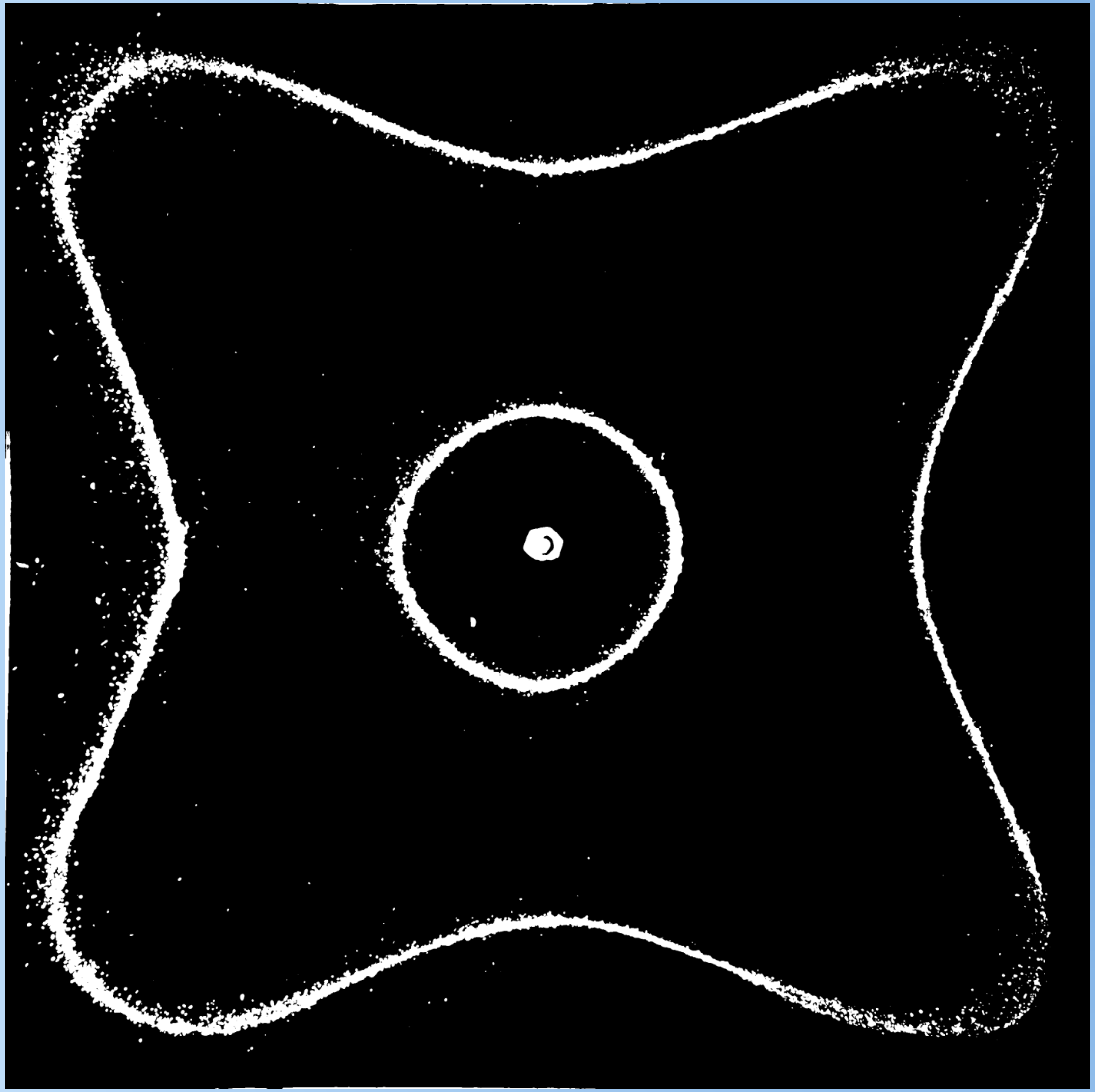
Binary Image

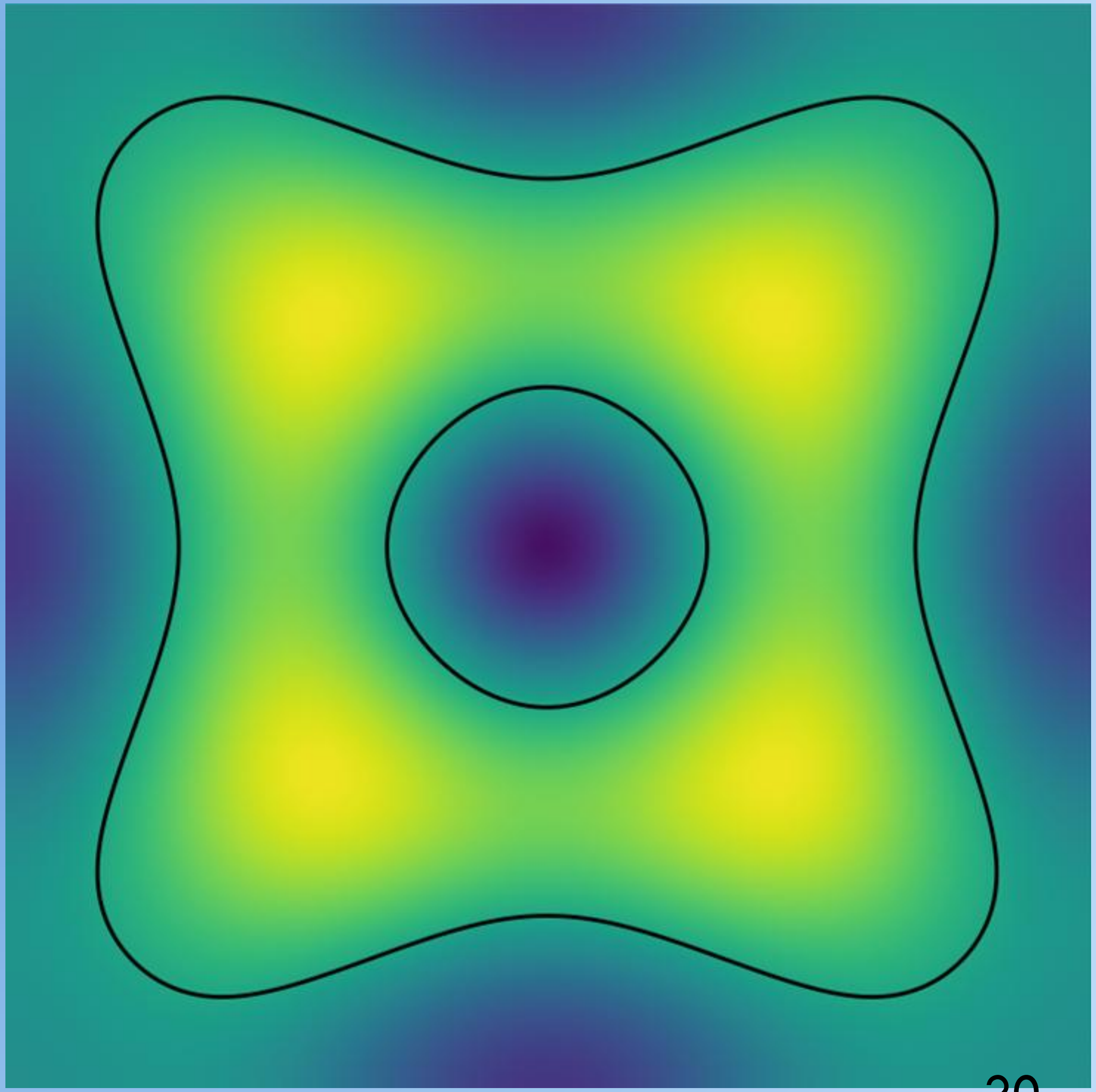


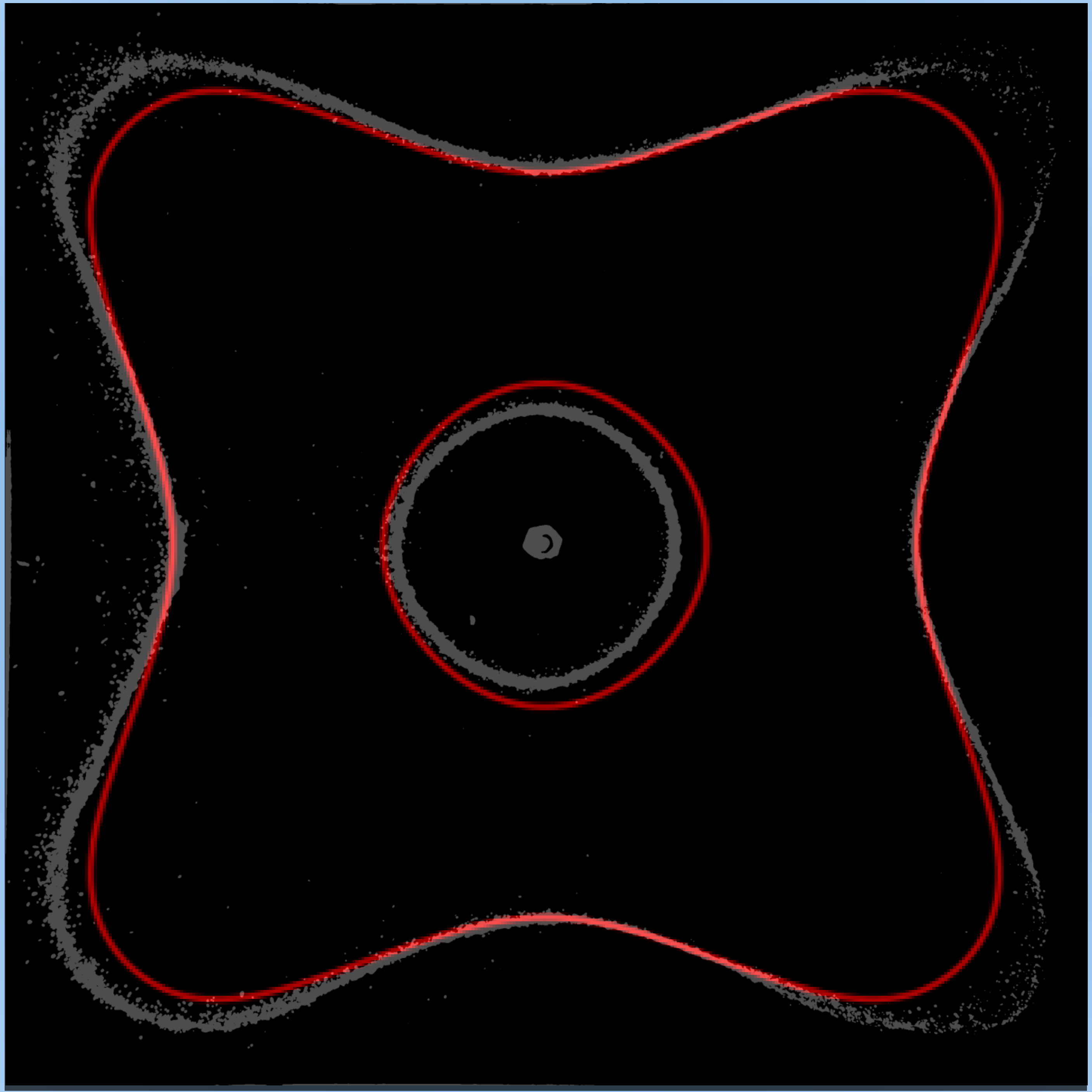
Composite

Binary Image

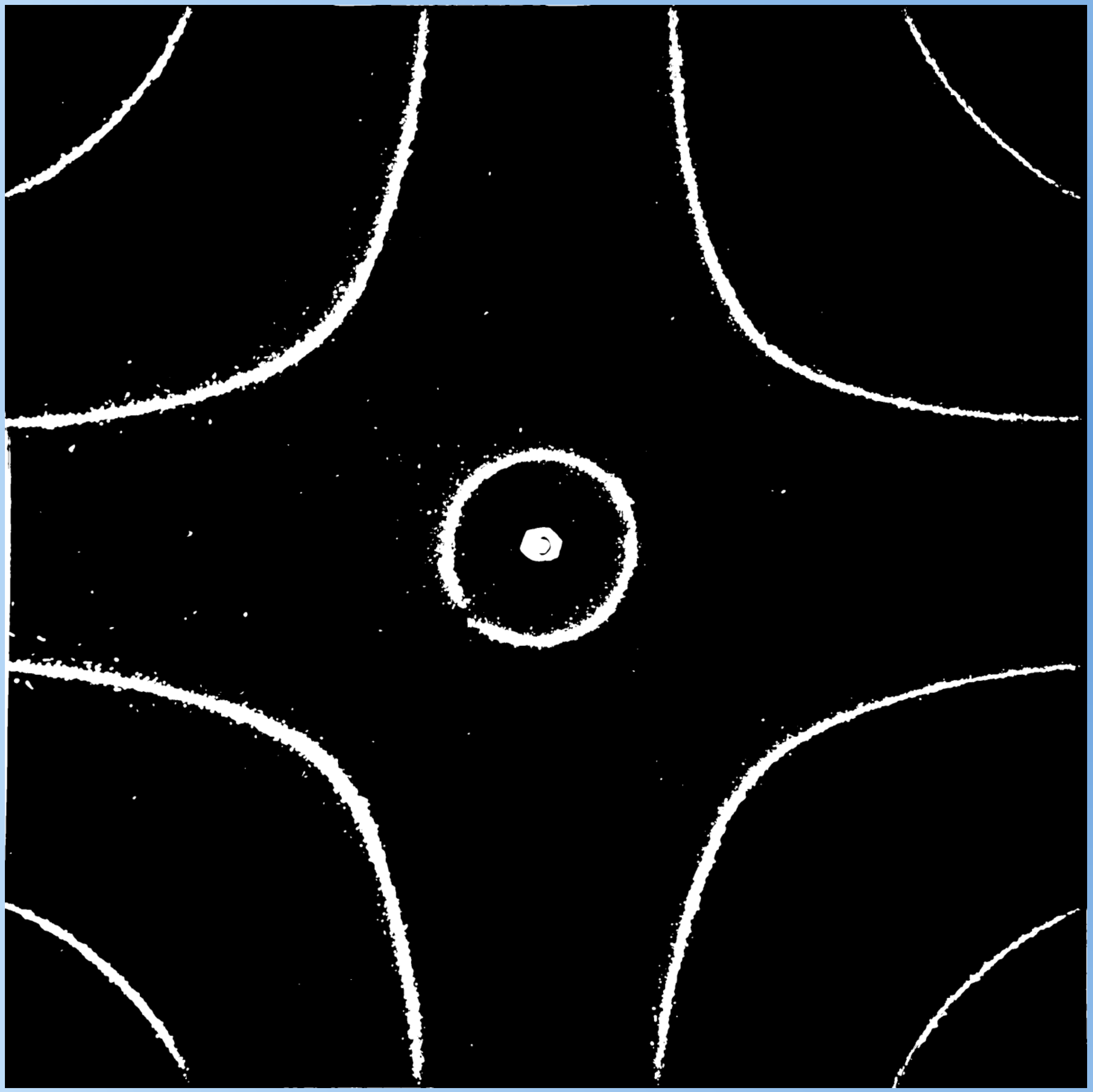
Comparison: 313 Hz, Center Driven

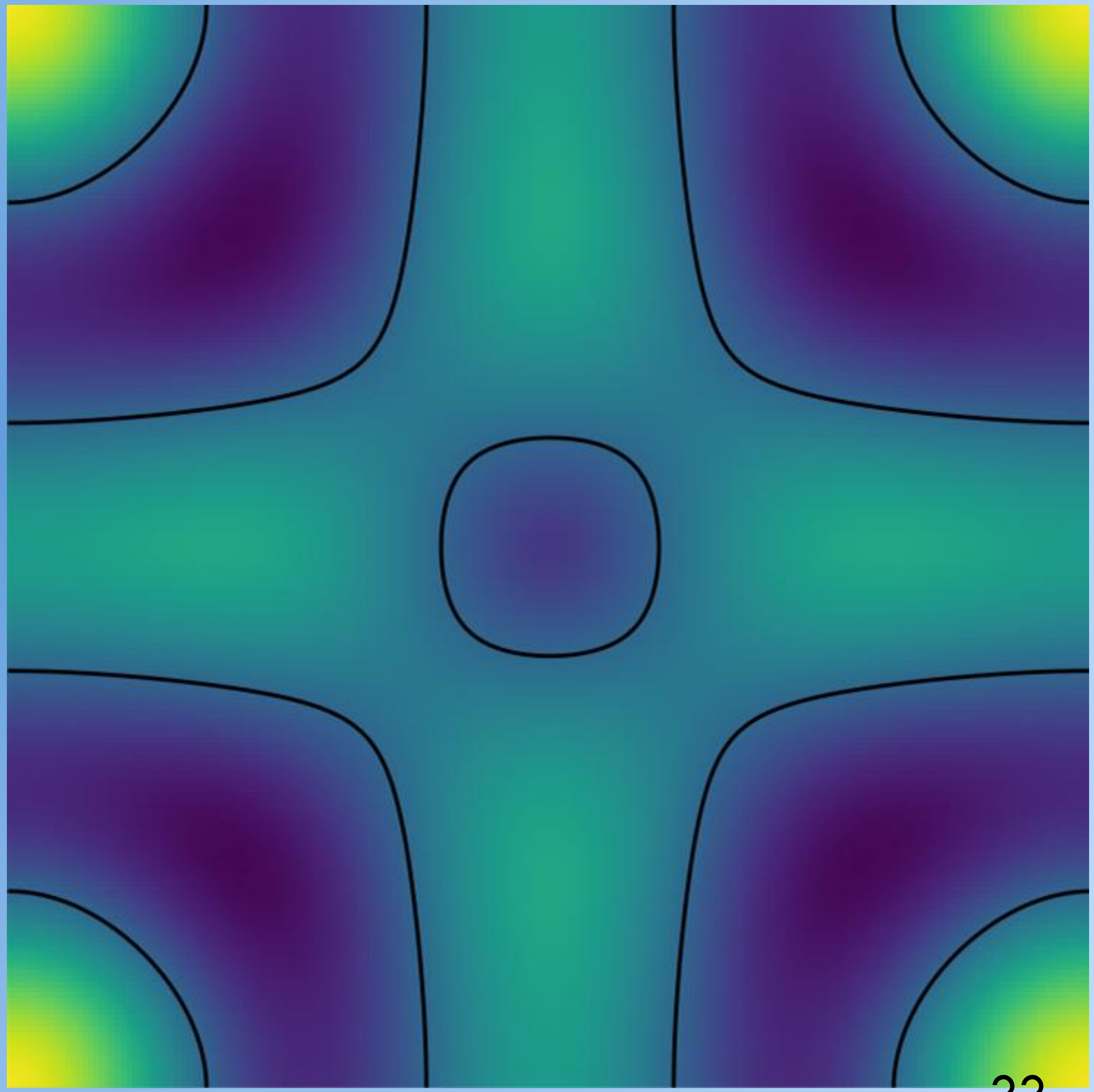


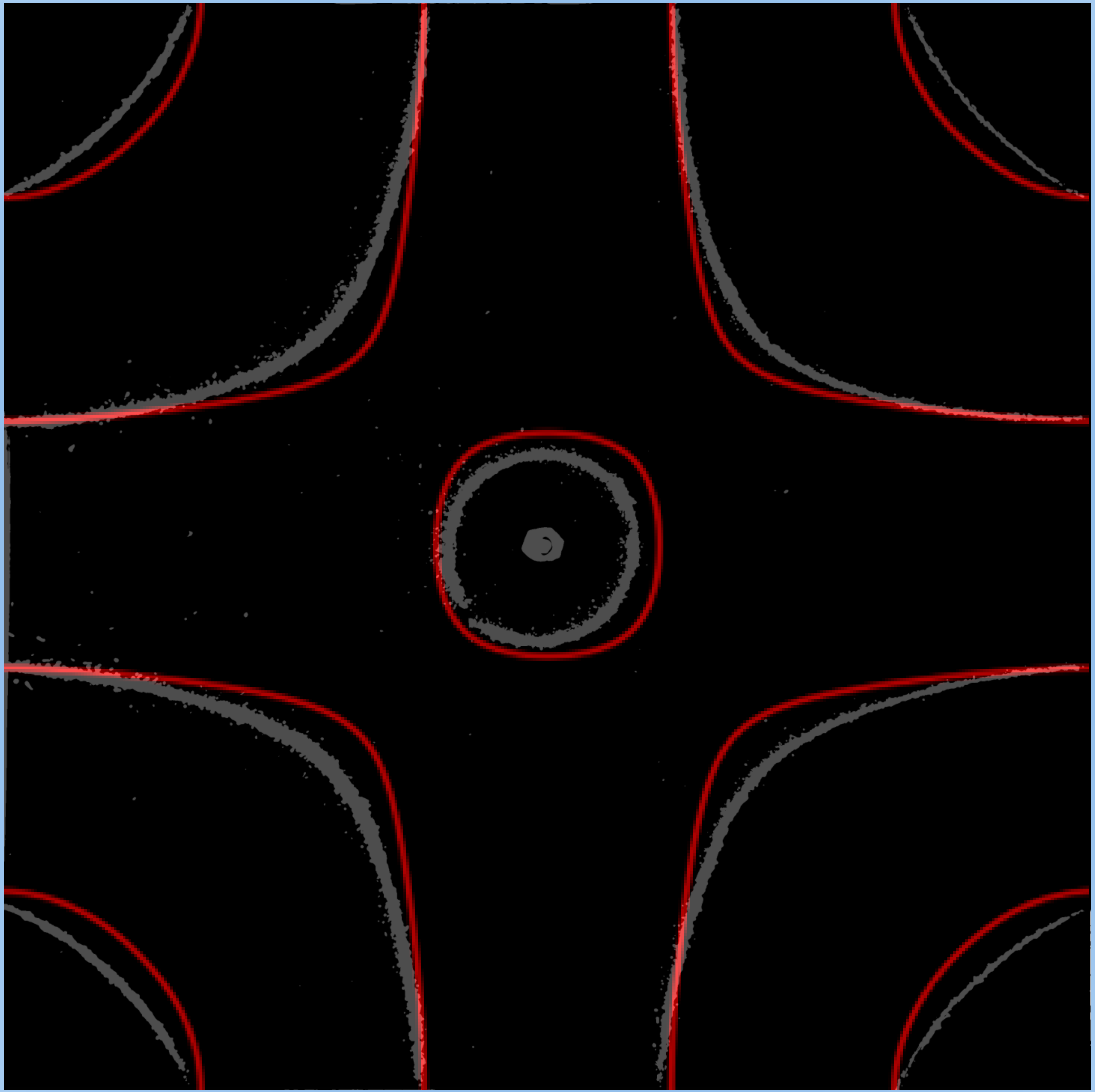




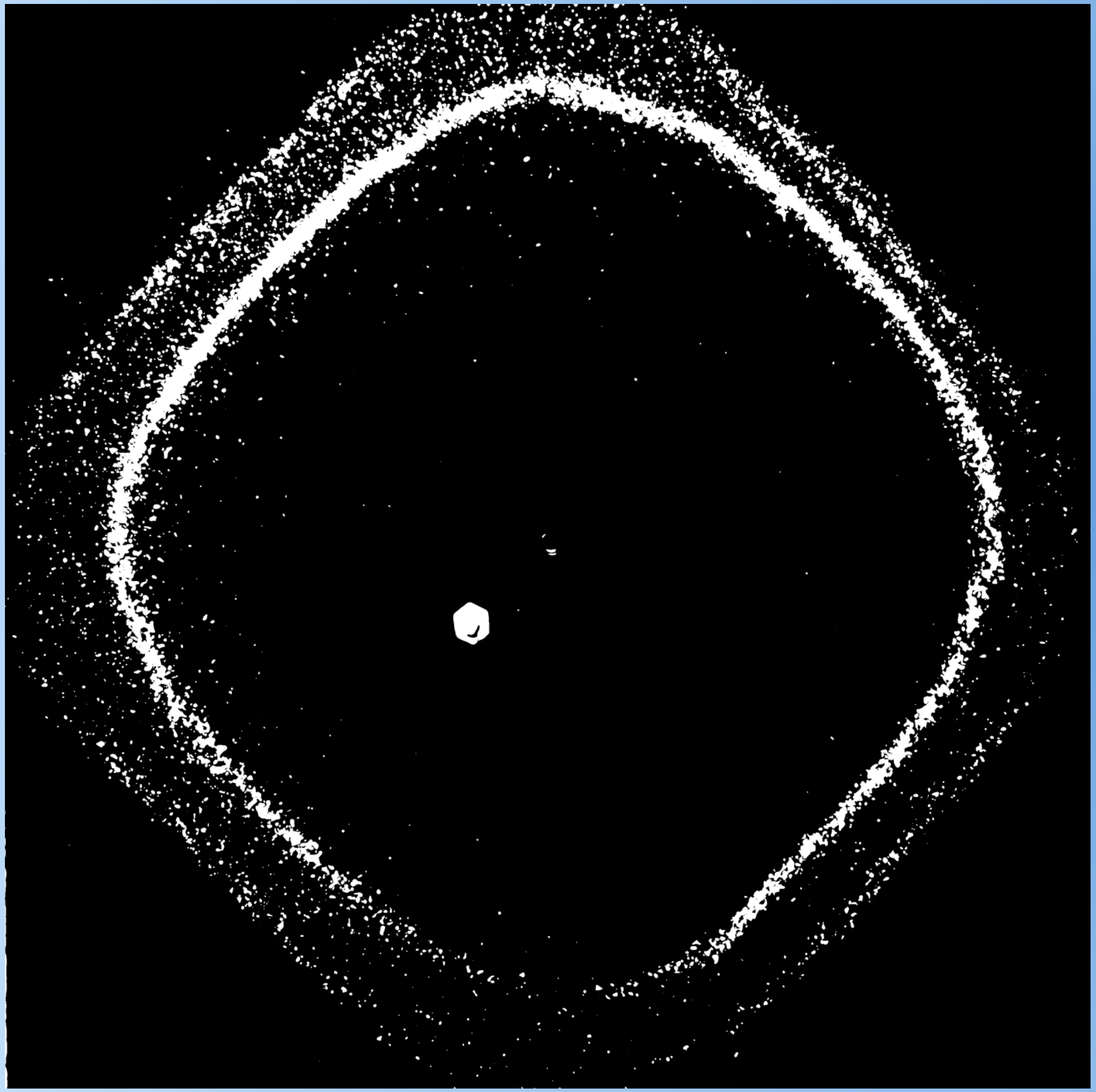
Comparison: 455 Hz, Center Driven

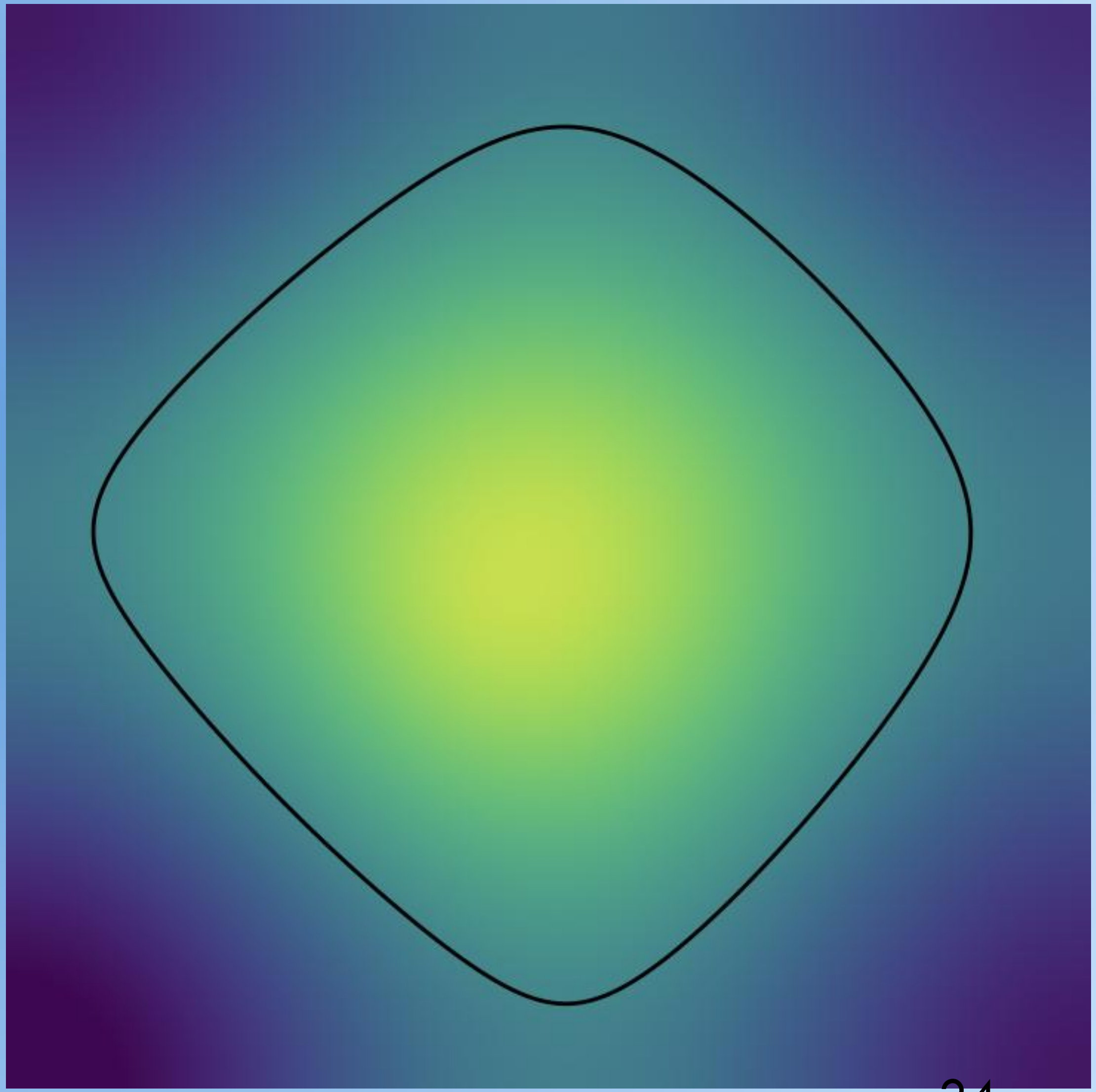


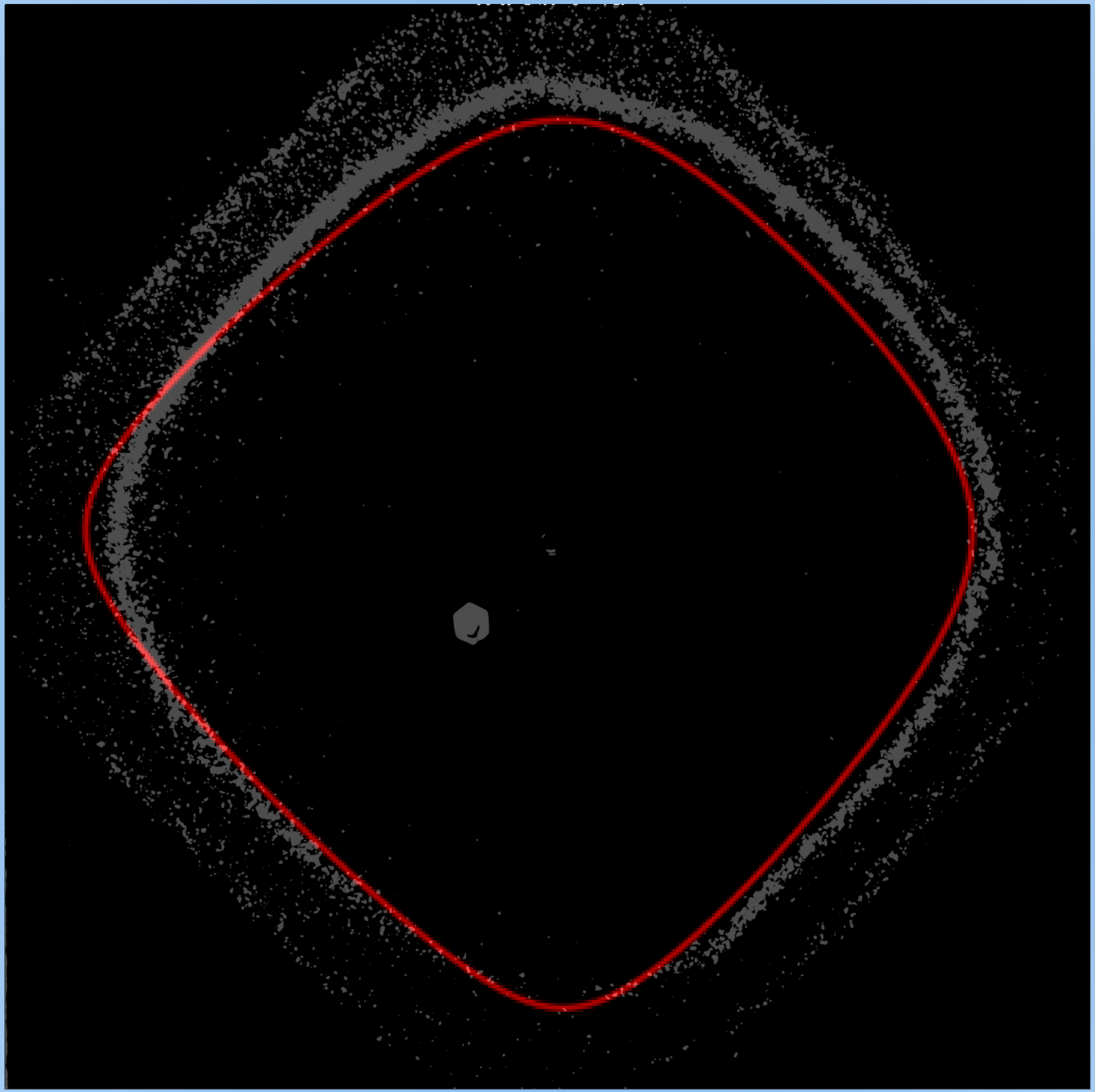




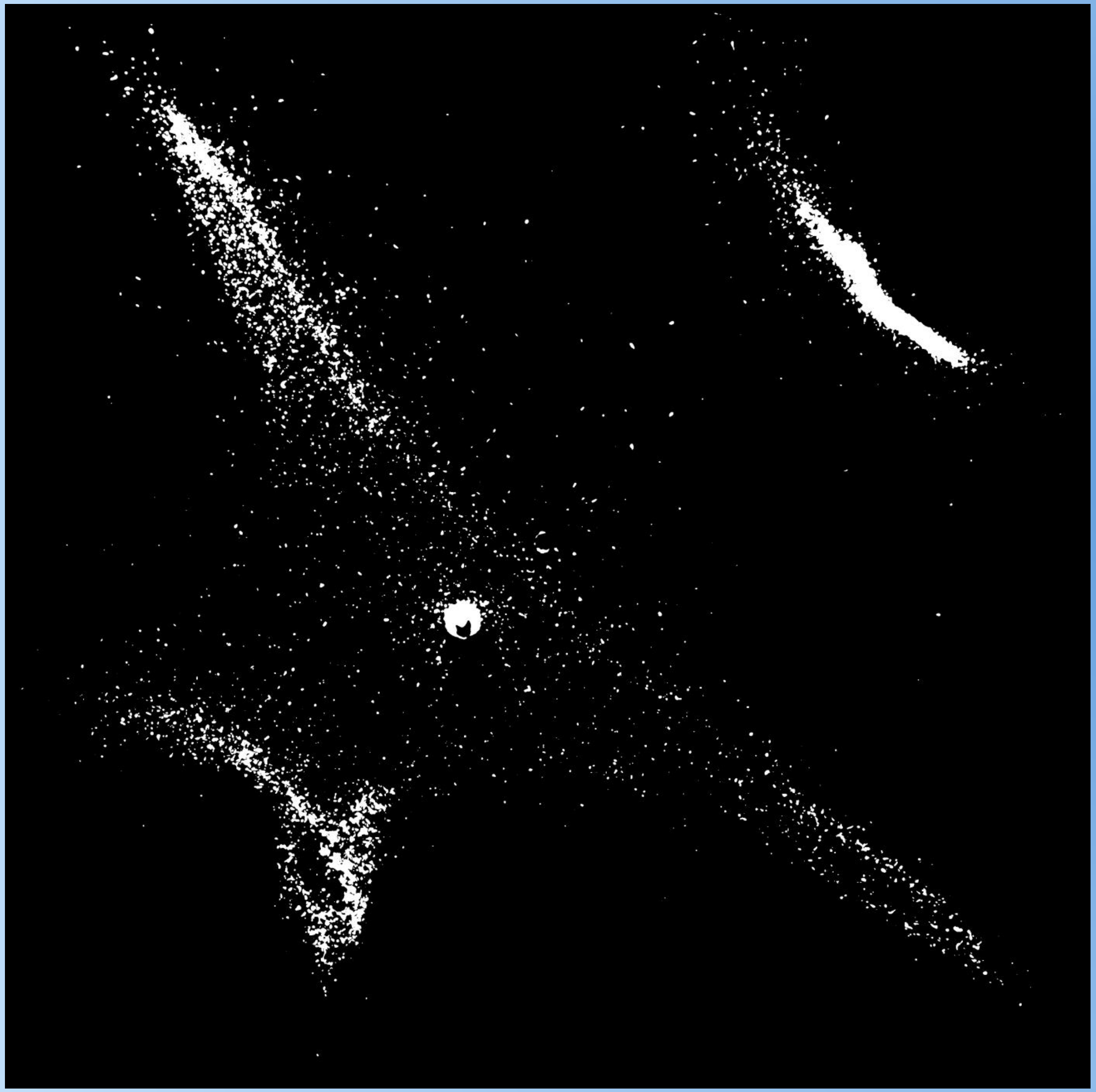
Comparison: 86 Hz, Off-Center Driven

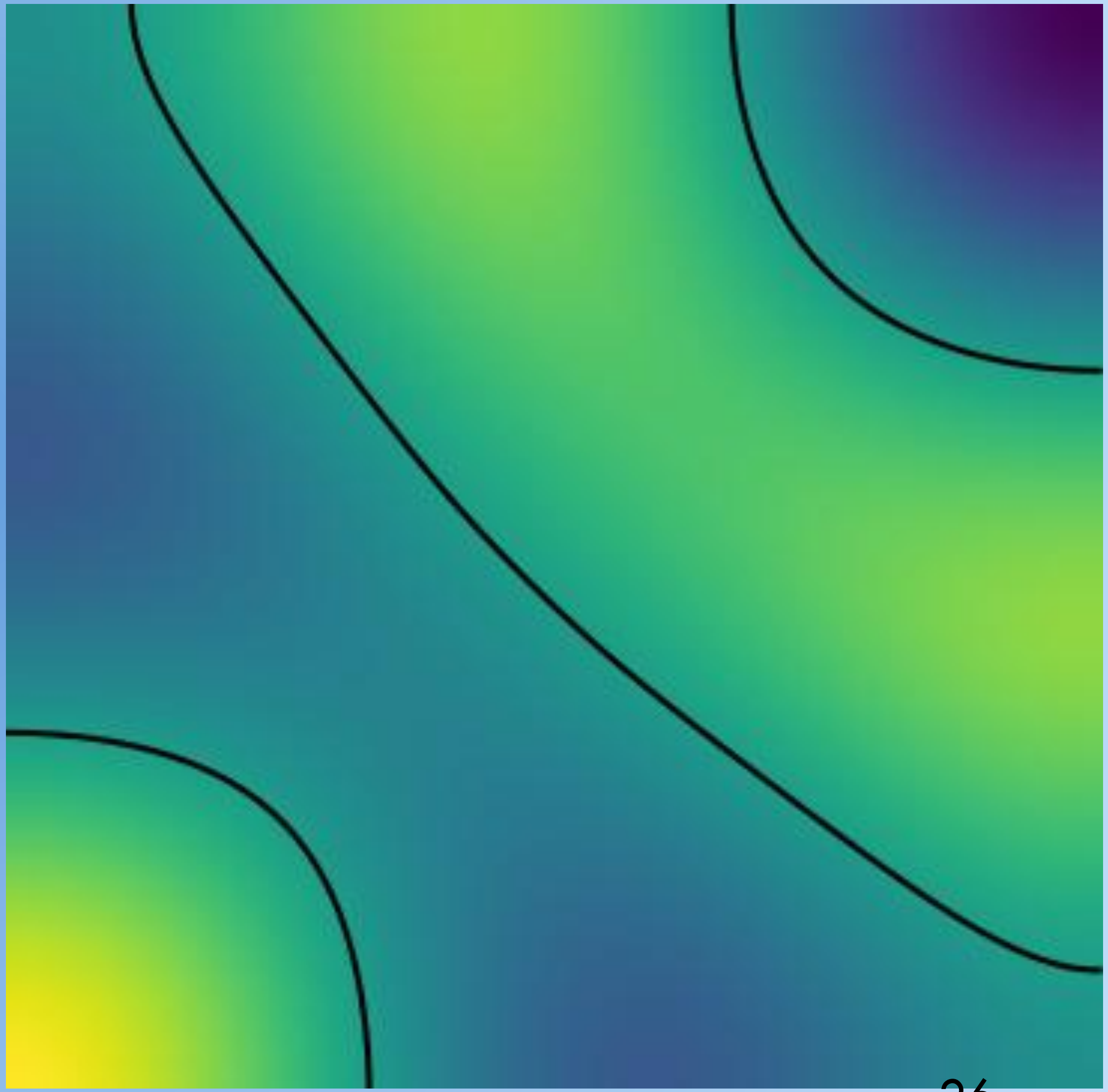


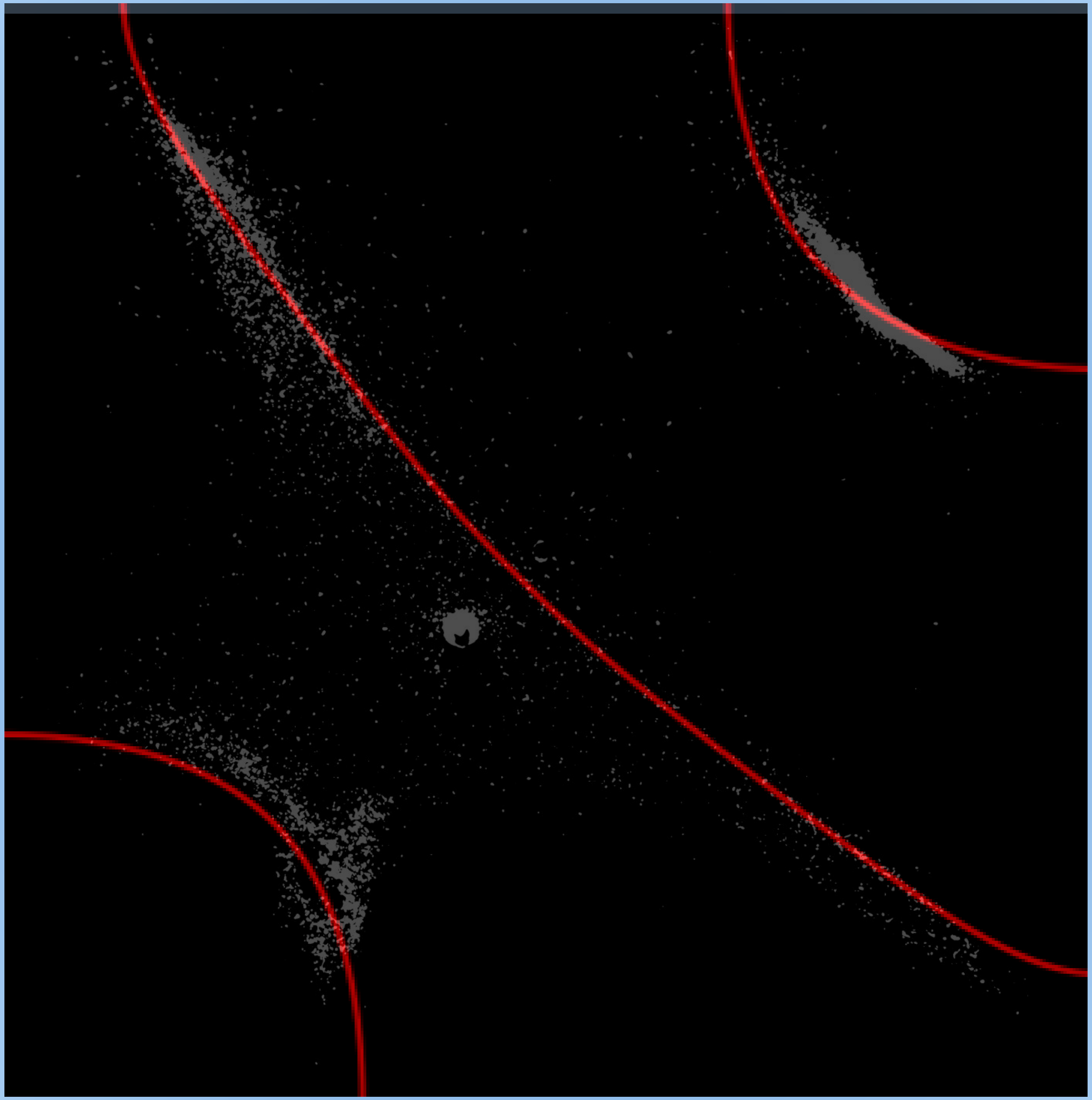




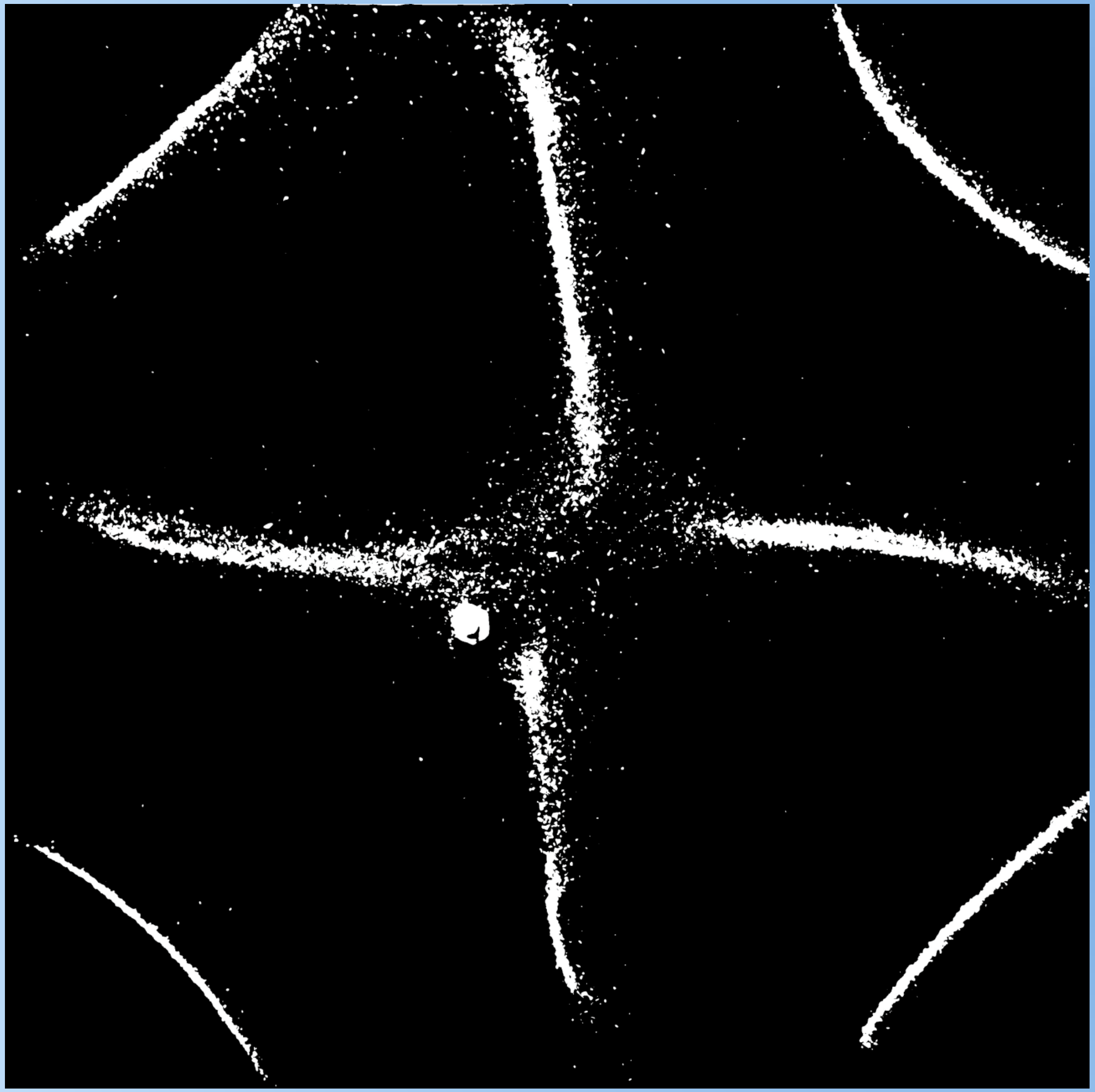
Comparison: 106 Hz, Off-Center Driven

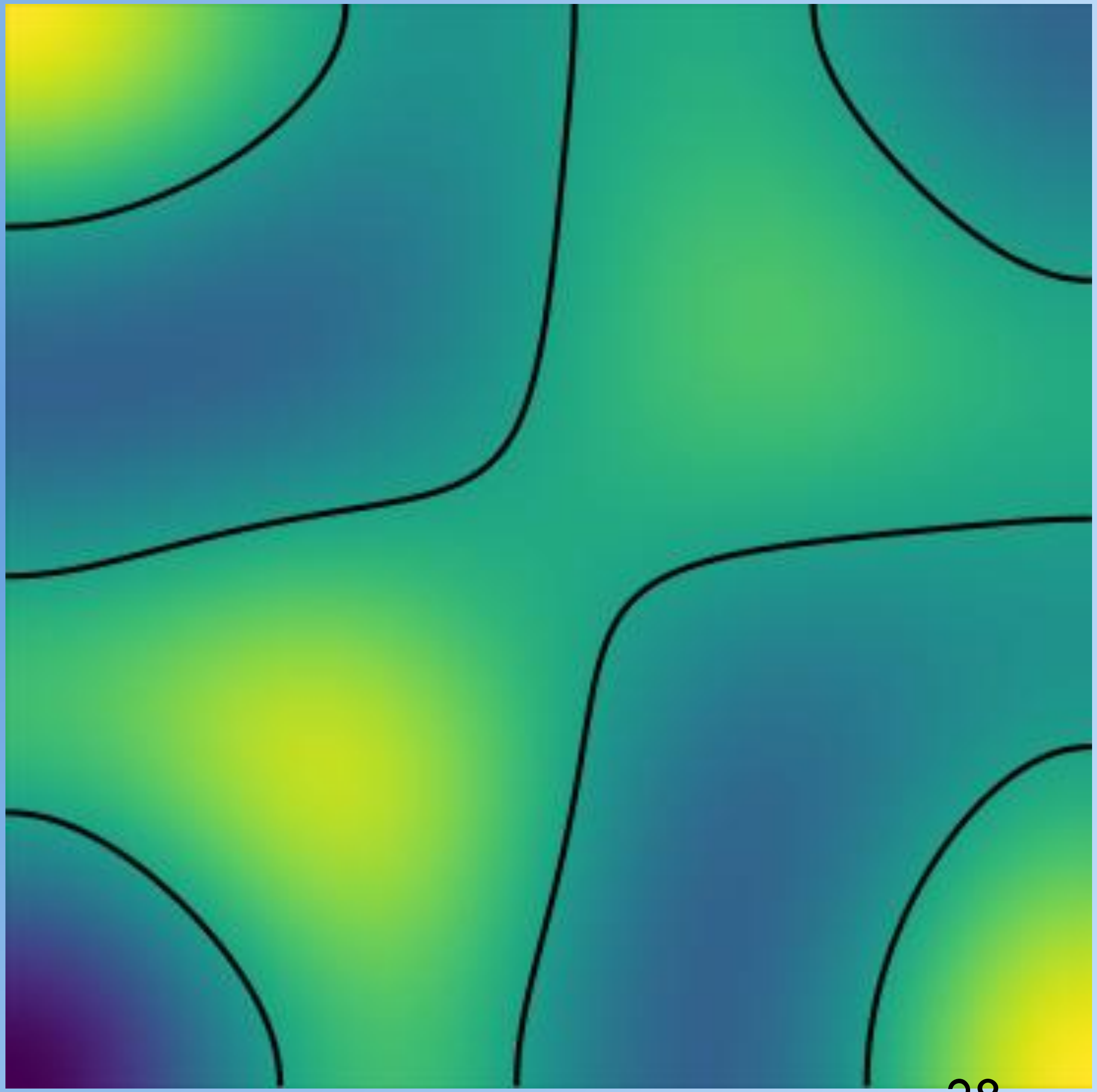


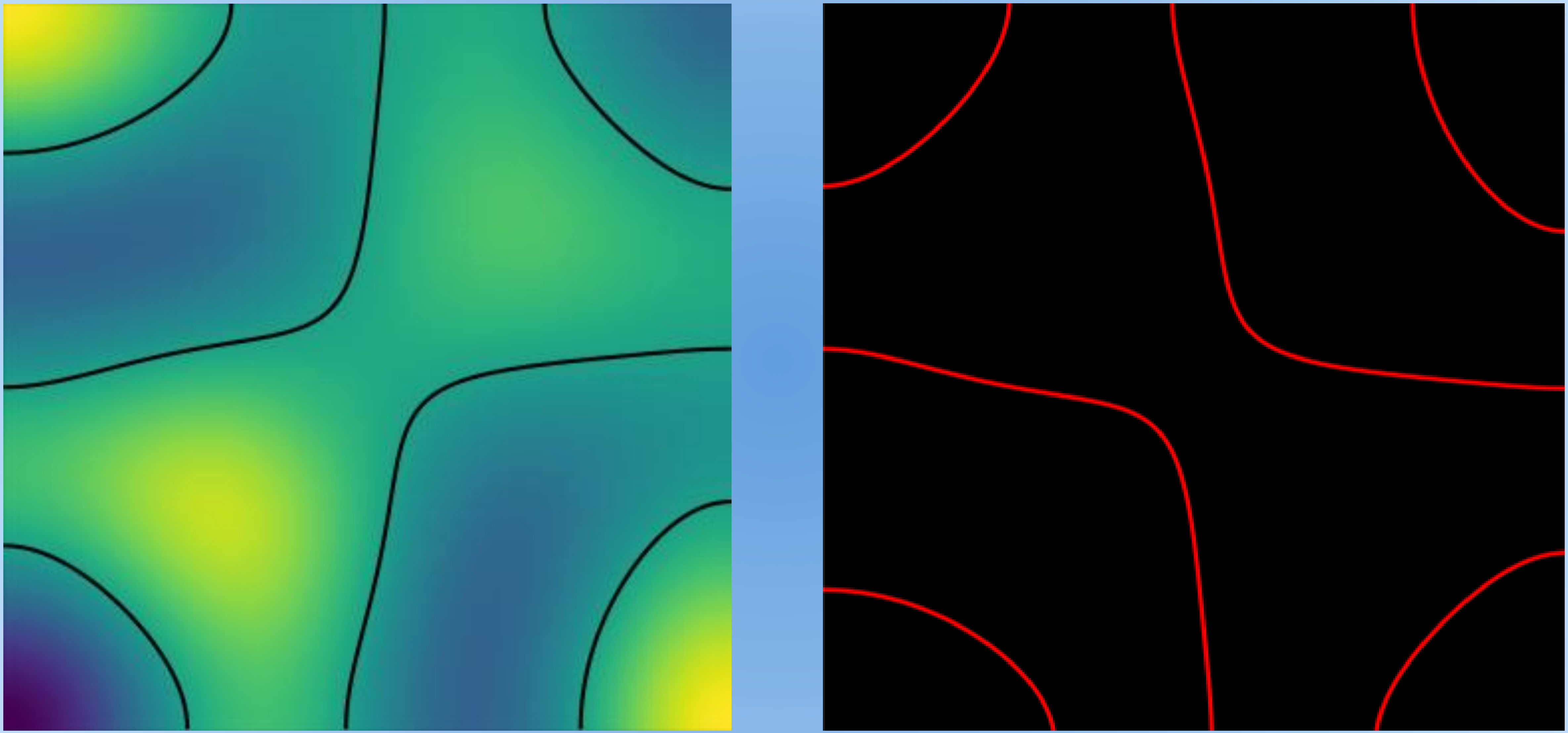


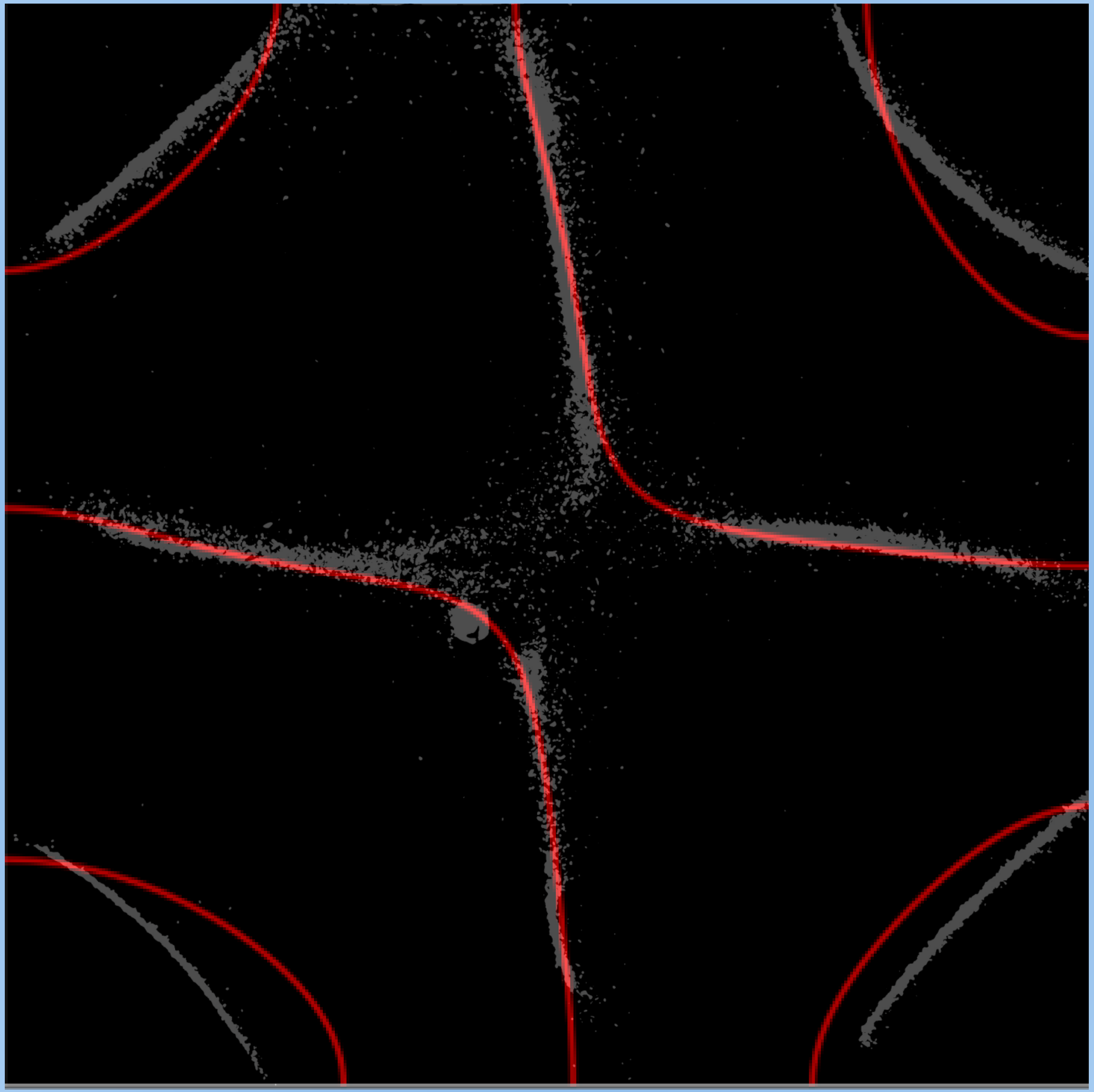


Comparison: 241 Hz, Off-Center Driven









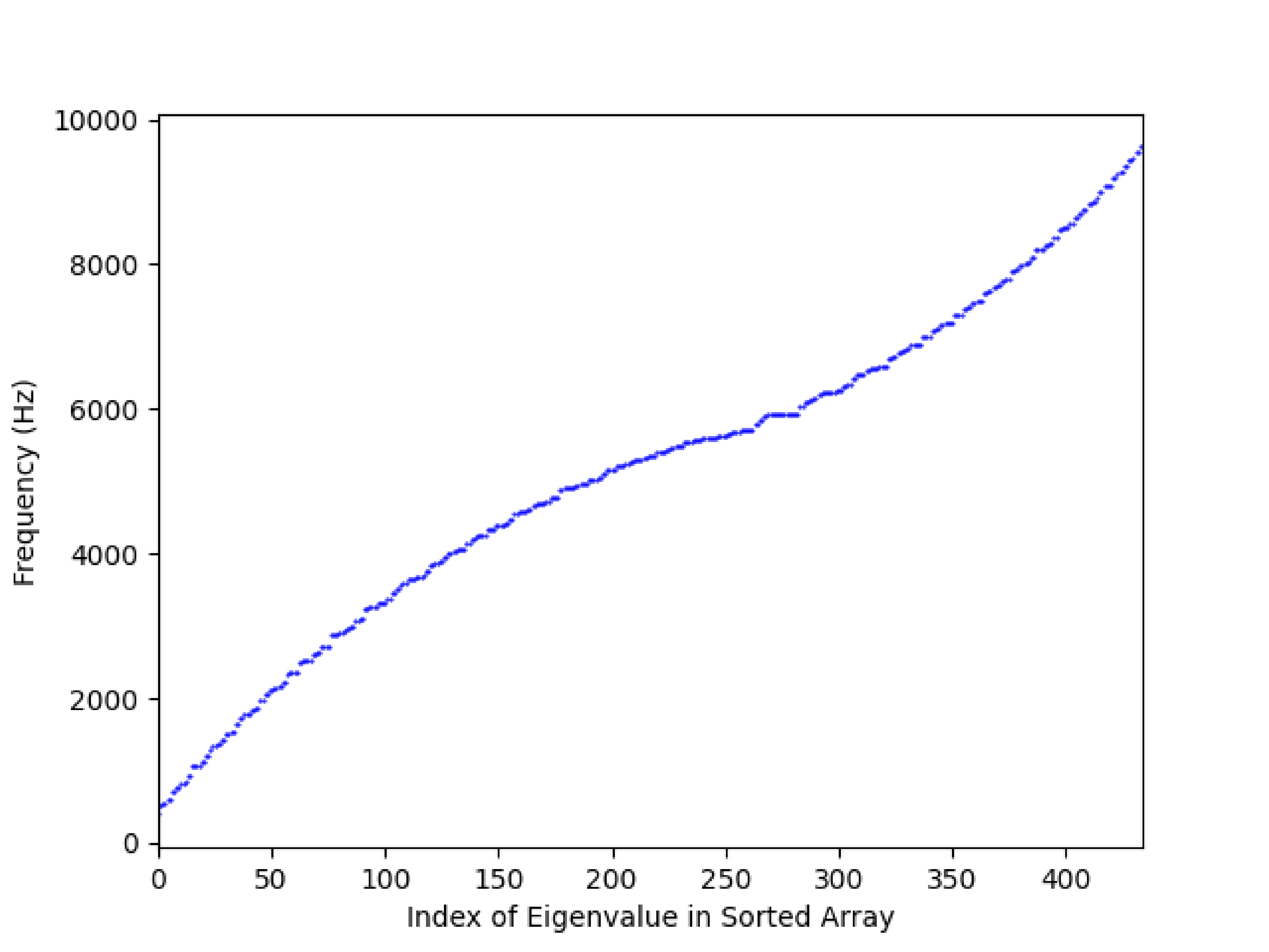
Eigenvalues

$$M\vec{u} + (\rho h)\omega^2\vec{u} = \vec{f}$$

$$M\vec{u} = -\lambda\,\vec{u}$$

$$\lambda = (\rho h)\omega^2$$

Many eigenvalues above our range of frequencies (60-1000 Hz)



Conclusions

- Several theory modes show a qualitative match
- Patterns that match well show "stretching"
- Use a plate that is flatter and has one hole
- There are a lot of nuances to solving this problem

Future Directions

- Solve homogenous Biharmonic equation
  - Solve for eigenvalues of operator matrix
  - Construct all possible harmonic mode patterns
- Solve by reducing to 2nd order equation
- Solve using the wave equation
- Develop resources to make similar projects more accessible

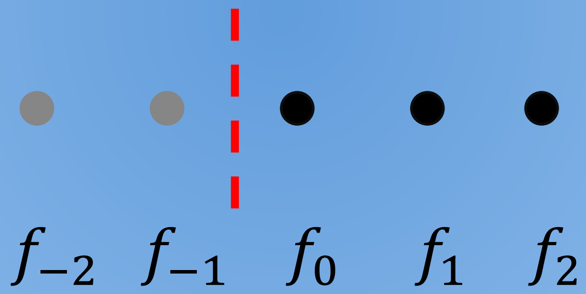
Questions?

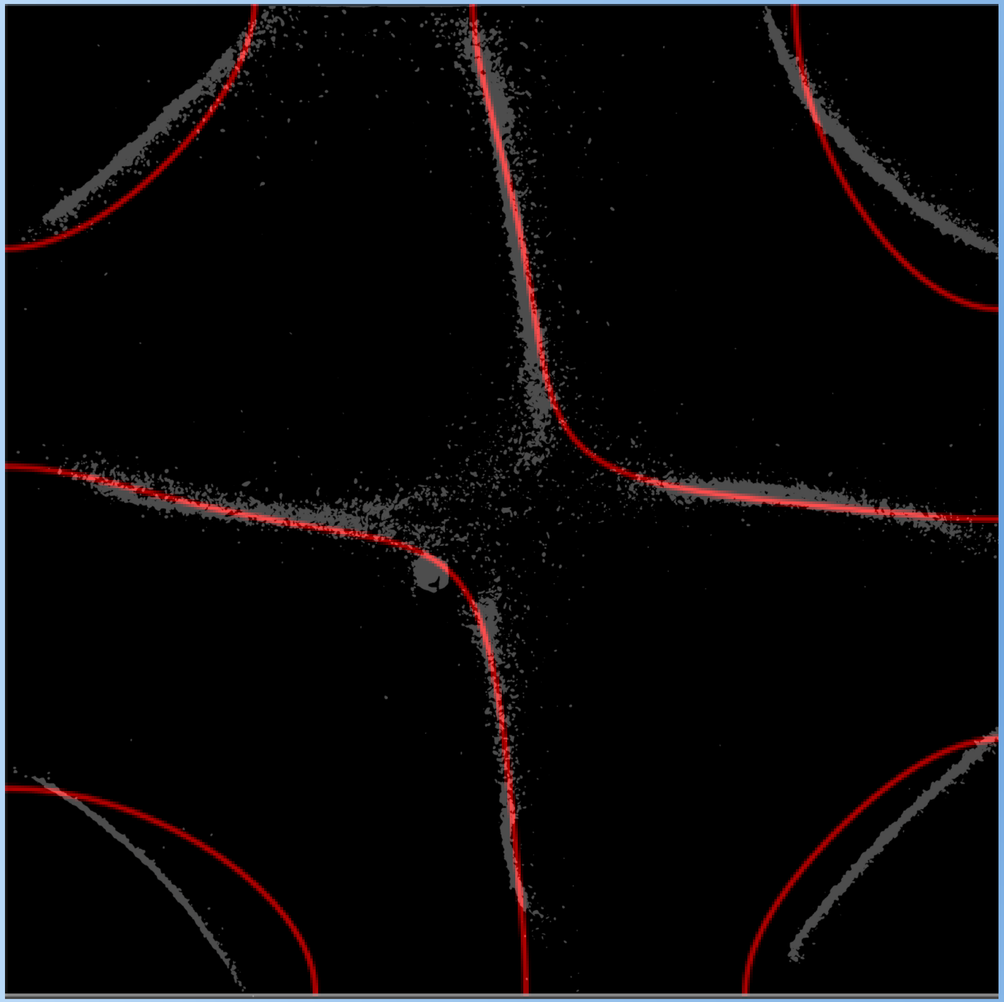
References

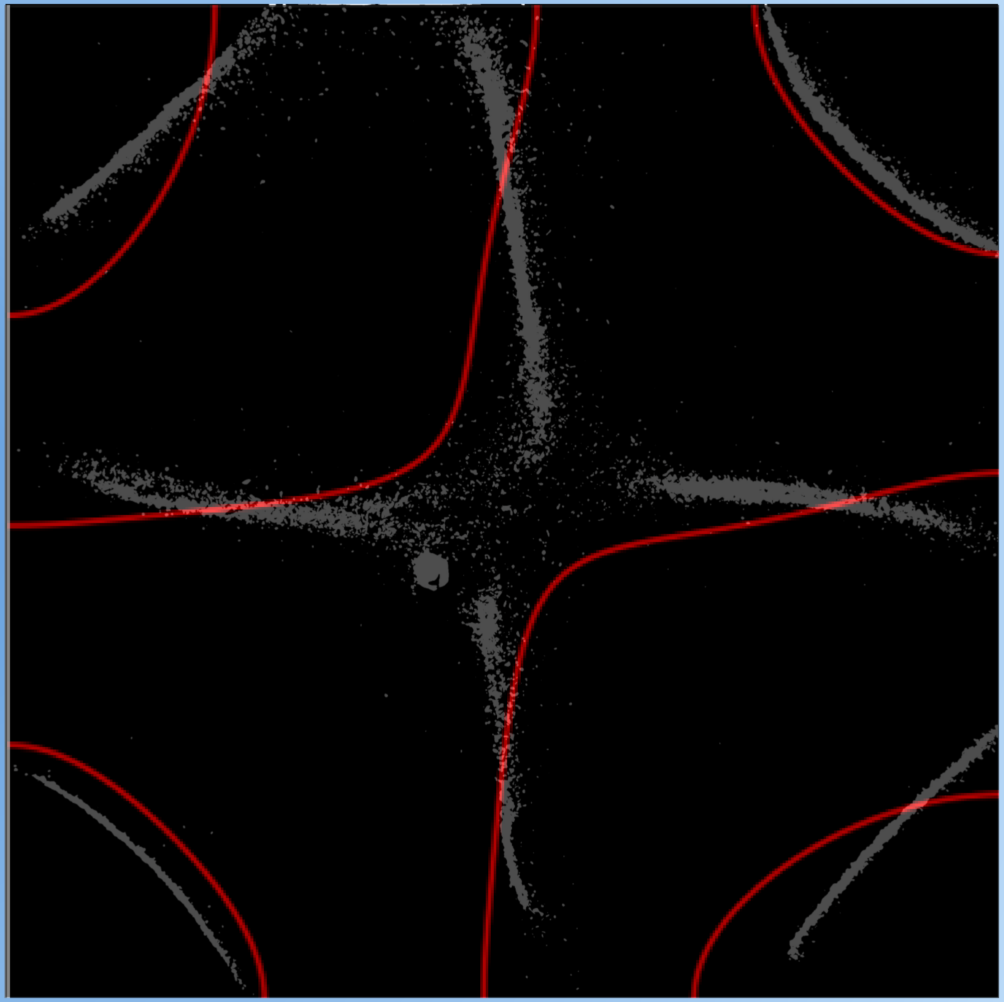
- D.T.A. Nguyen, L. Lin, H. Ji, *Stable and Accurate Numerical Methods for Generalized Kirchhoff-Love Plates*
- S. Kanchanapalli, *Mathematically Modeling Chladni Plate Patterns*
- X. Meng, G. Yang, Y. Yin, R. Xiao, *Finger Vein Recognition Based on Local Directional Code* (grayscaling)

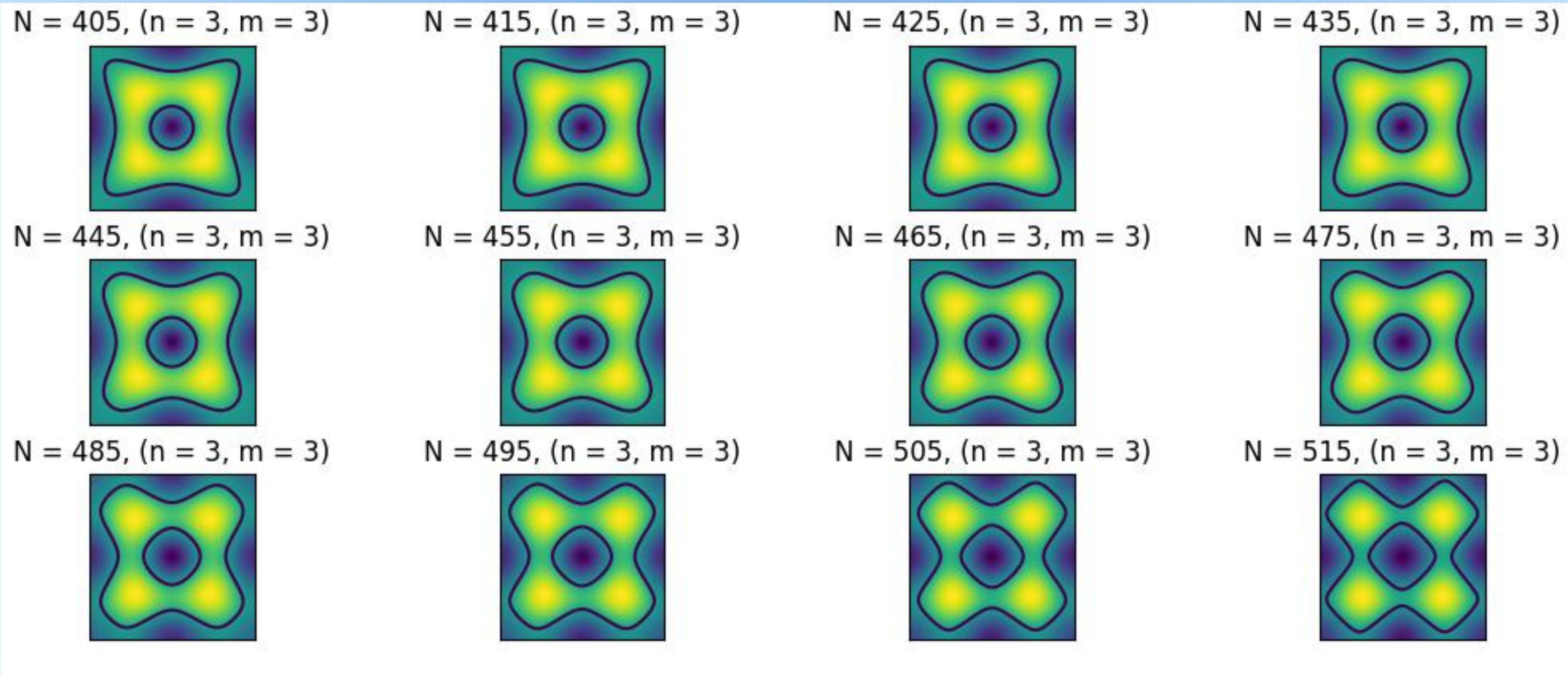
Ghost Points

$$\frac{\partial f}{\partial x} = 0 \Rightarrow \frac{f(x+h) - f(x-h)}{2h} = 0$$





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